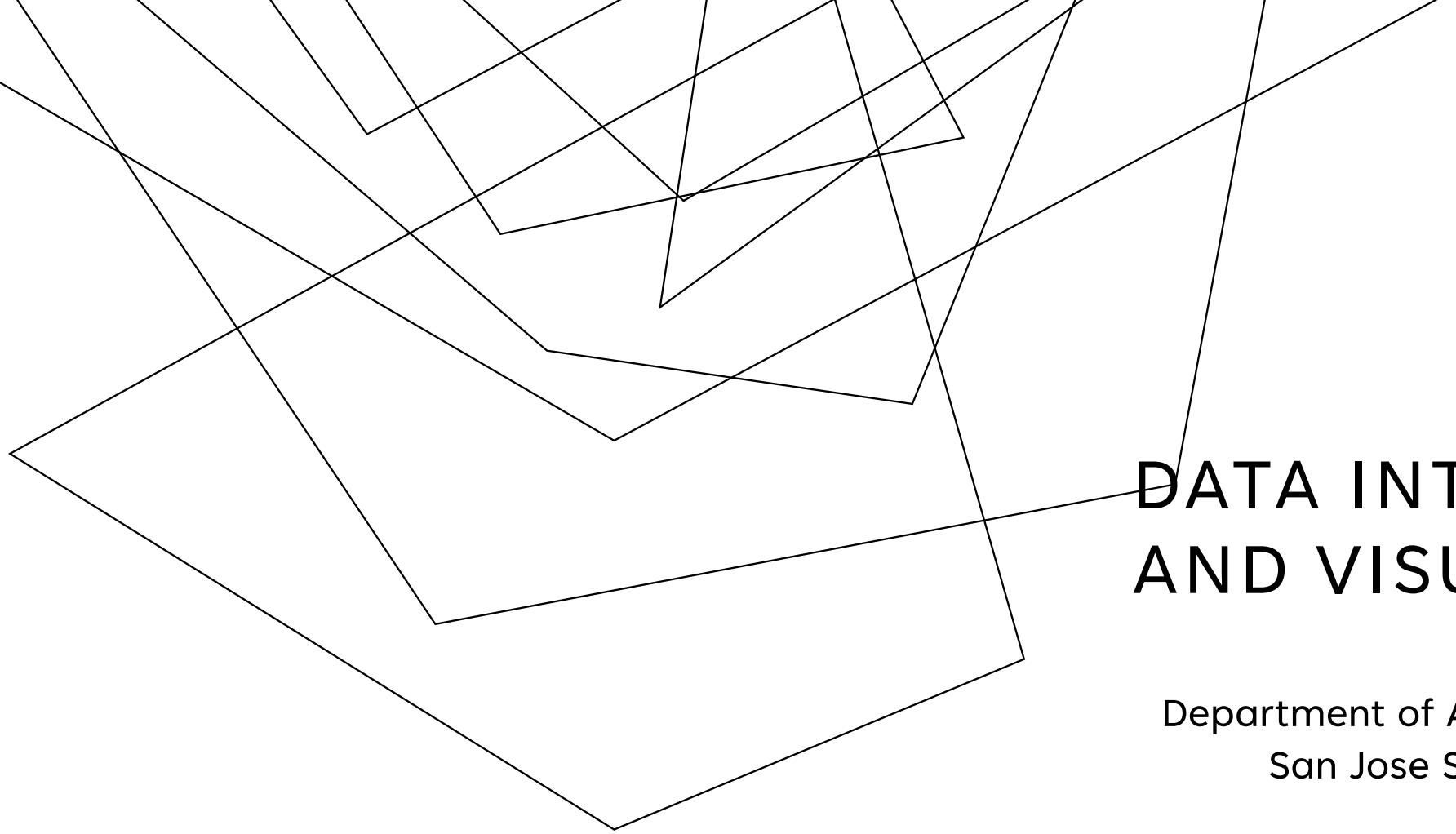


DATA INTELLIGENCE AND VISUALIZATION

Department of Applied Data Science
San Jose State University

Instructor: Dr. Guannan Liu

September 18

Fall 2026

TODAY'S LECTURE

- **Review Last Session**
- EDA Task
- Uncertainty Plot
- Visualizing Multiple Distributions

The Shape of an EDA

Structure – Clean – Univariate – Bivariate – Multivariate – Decisions

Act	Question it answers	Charts
1 Structure and cleaning	What is actually in this file, and what is broken?	console output, missing-rate bar
2 Univariate — categorical	What kind of cars are these?	bar, grouped bar, dot plot, heatmap
3 Univariate — numeric	What shape is each variable, and the target above all?	histogram, density, box, violin, QQ
4 Bivariate	What moves with price?	scatter, correlation, box by group
5 Multivariate	What is really driving it, once you hold something constant?	heatmap, pairplot, faceted scatter

The sixth stage produces a **table of decisions** is what the next person on your project actually needs.

Structure and Cleaning

Structure – Clean – Univariate – Bivariate – Multivariate – Decisions

- **Every EDA starts the same way**
  - df = pd.read_csv('used_cars_data.csv')
  - df.head(5)/df.tail(5) — what does one record look like?
  - df.shape — 7,253 rows x 14 columns
  - df.info() — types and non-null counts
  - df.describe() — the five-number summary of every numeric column

The Output of an EDA Is a Decision Table

Structure – Clean – Univariate – Bivariate – Multivariate – Decisions

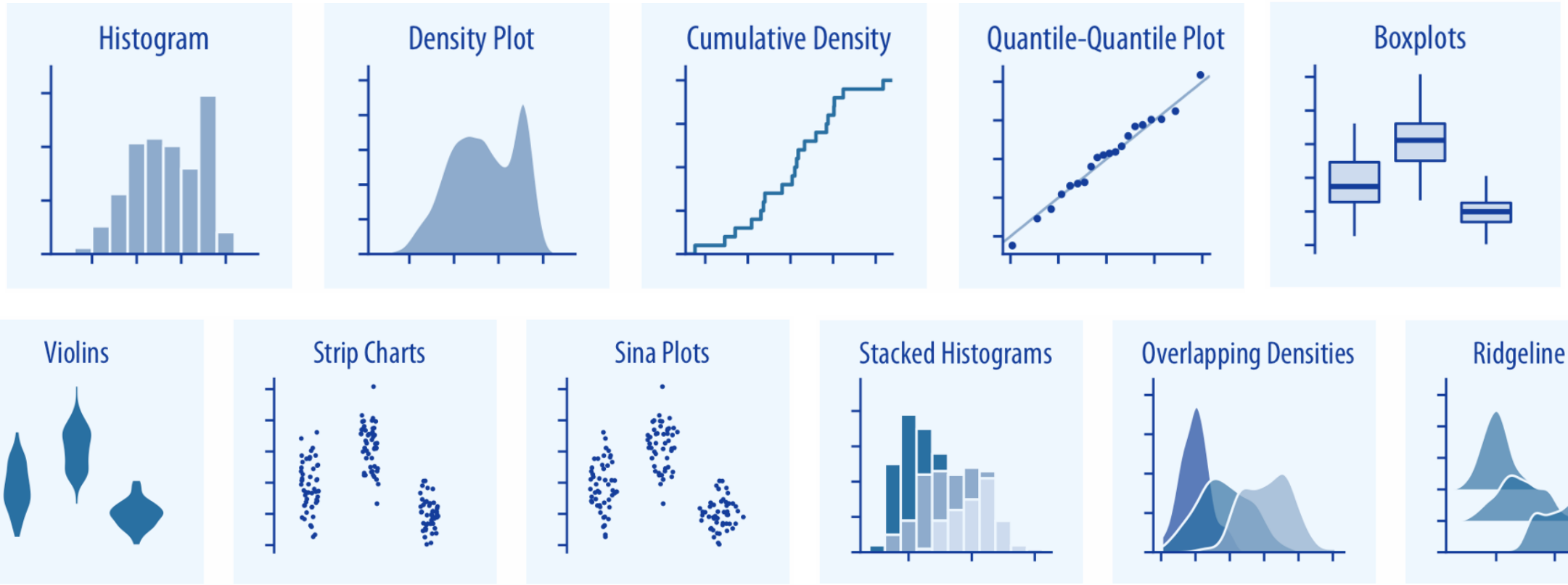
Column / issue	Decision	Because
S.No.	drop the column	row number; it also disabled duplicated()
Mileage, Engine, Power	strip units, cast to float	describe() was silently skipping them
New_Price rows in 'Cr'	convert Cr to Lakh (x100)	1 Cr = 100 Lakh; 20 rows
Power == 'null bhp'	treat as missing	129 rows; true missing rate 0.6% -> 2.4%
Mileage == 0 / Seats == 0	treat as missing	81 rows and 1 row; both impossible
duplicate row	drop	only visible once S.No. was removed
New_Price	drop the column	86% missing
Price	drop rows where missing	it is the target; never impute a target

Directory of Visualizations: Amounts



Source: Fundamentals of Data Visualization,
Claus O. Wilke, O'Reilly Media 2019

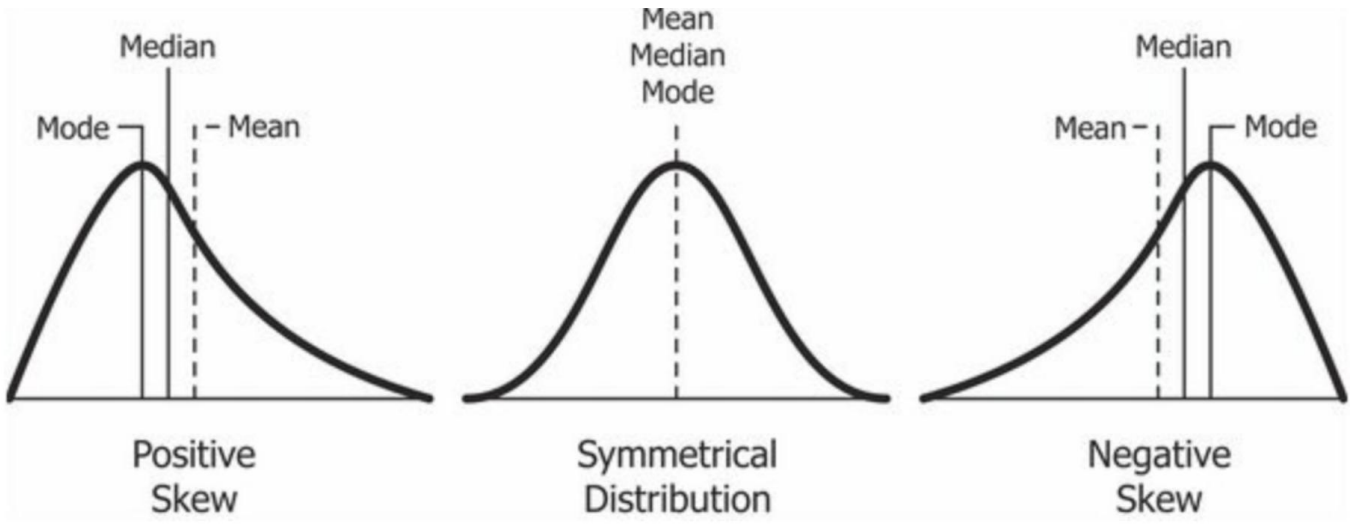
Directory of Visualizations: Distribution



Source: Fundamentals of Data Visualization,
Claus O. Wilke, O'Reilly Media 2019

Distribution Analysis

- **Shape:**
  - **Skewness**: Is the histogram symmetrical or skewed? (**function: stats.skew()**)
    - **Symmetrical**: Looks the same on both sides of the center; Mean = Median = Mode
    - **Skewed Right (Positive Skew)**: Tail is longer on the right side; Mean > Median > Mode
    - **Skewed Left (Negative Skew)**: Tail is longer on the left side; Mean < Median < Mode



EDA: Univariate Analysis

```
labels = ['kilometer driven (km)', 'price', 'mileage (kmpl)',
          'new price (Lakh)', 'engine (CC)', 'power (bhp)',
         'car age (years)']

bin_size = 30

for i, column in enumerate(numerical_df.columns):
    plt.figure(figsize=(8, 3))
    plt.subplot(1, 3, 1)
    plt.hist(numerical_df[column].dropna(), bins=bin_size, color='skyblue', edgecolor='black')
    plt.xlabel(labels[i] if i < len(labels) else column)
    plt.ylabel('Frequency')
    plt.grid(True, axis='y', alpha=0.5)
    plt.subplot(1, 3, 2)
    sns.kdeplot(numerical_df[column].dropna(), fill=True, color='orange')
    plt.xlim(left=0)
    plt.xlabel(labels[i] if i < len(labels) else column)
    plt.ylabel('Density')
    plt.grid(True, axis='y', alpha=0.5)
    plt.subplot(1, 3, 3)
    plt.boxplot(numerical_df[column].dropna(), vert=True, patch_artist=True,
                boxprops=dict(facecolor='lightgreen', color='black'))
    plt.ylabel(labels[i] if i < len(labels) else column)
    plt.grid(True, axis='y', alpha=0.5)

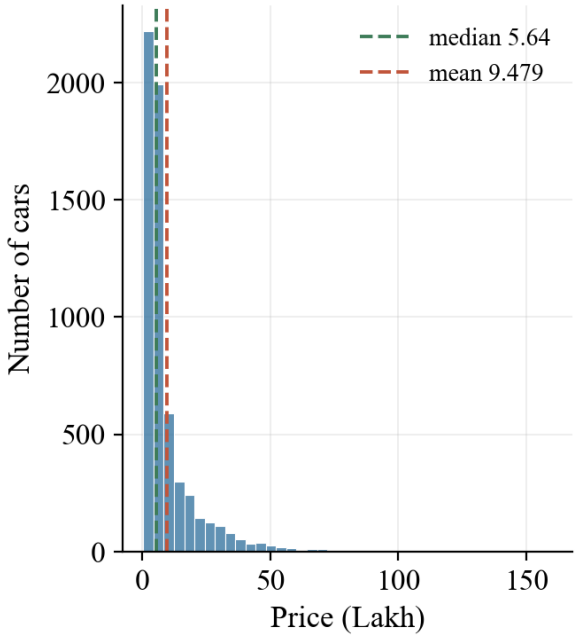
    plt.tight_layout()
    plt.show()
```

Univariate: Price (the target)

Price (the target) — n = 6,019, 1,233 missing

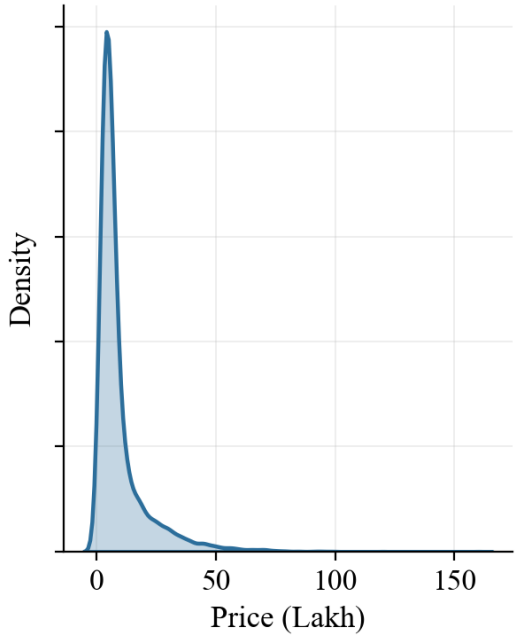
1 · Histogram

the mean sits 68% above the median



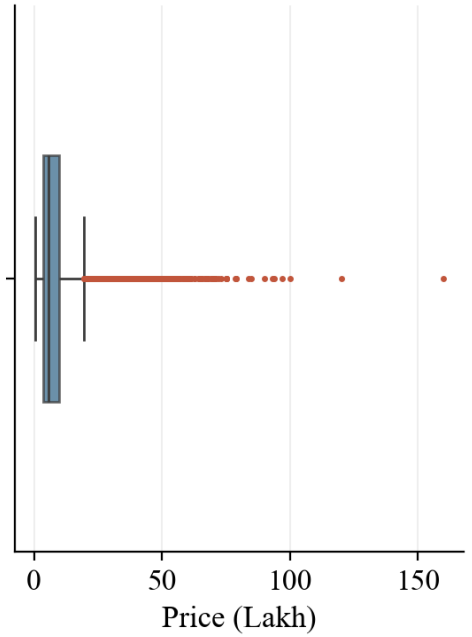
2 · Density

skew +3.33 (right tail)



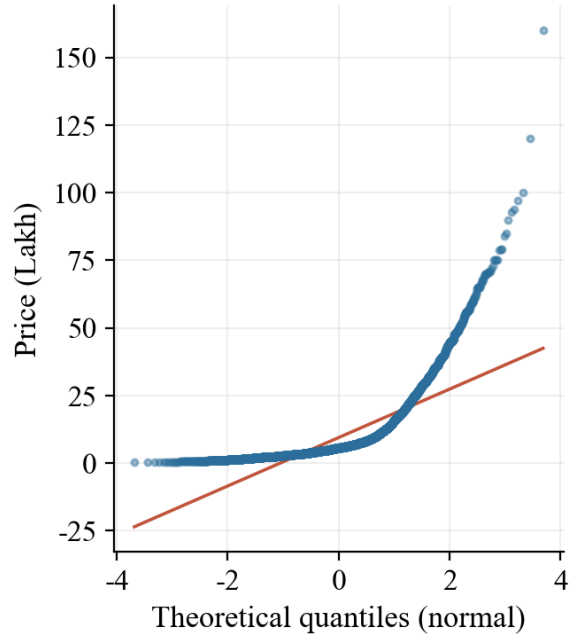
3 · Box plot

718 rows past the fences (11.9%)



4 · QQ plot

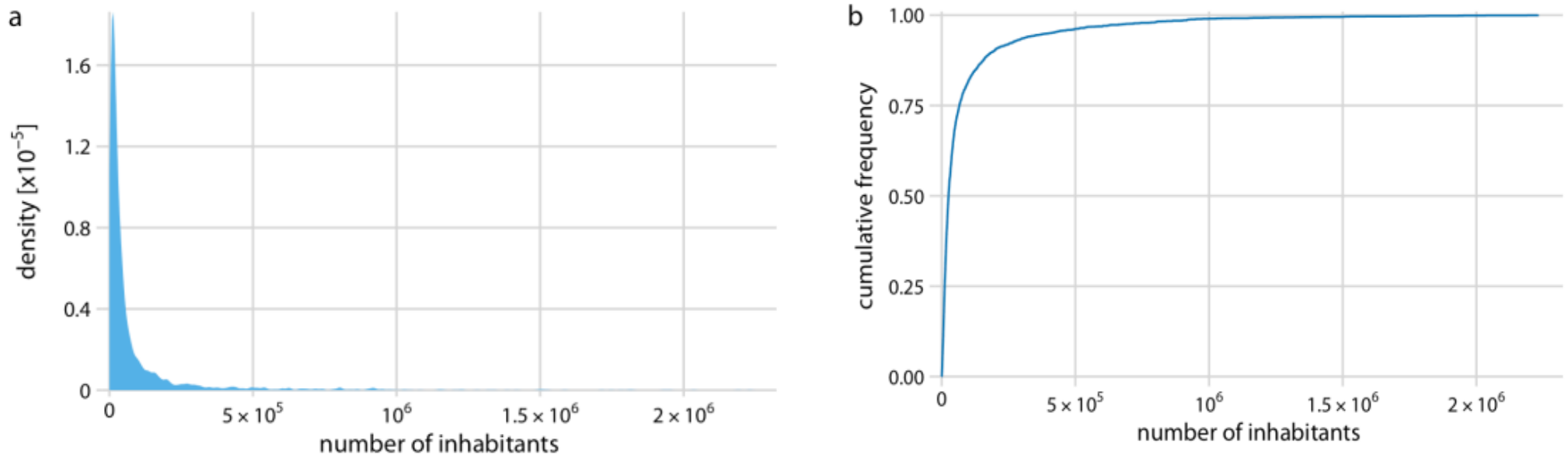
R^2 = 0.640 (bends away)



The target is not normal. Skew +3.33, 718 rows past the upper fence, and a QQ plot that leaves the line at both ends. This is the evidence behind log(Price) later in the act.

Highly Skewed Distribution

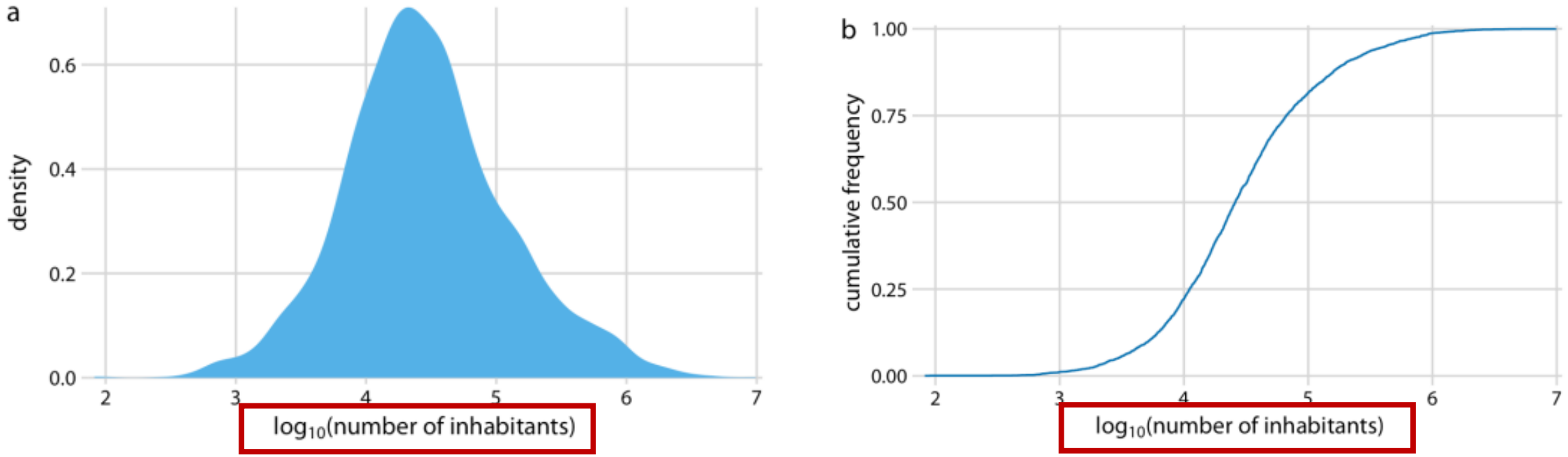
- The density plot (Figure 8-4a) shows a sharp peak right at 0.
- The ECDF (Figure 8-4b) shows a rapid rise near 0



- Figure 8-4. Distribution of the number of inhabitants in US counties. (a) Density plot. (b) Empirical cumulative distribution function. Data source: 2010 US Decennial Census.

Highly Skewed Distribution

- The density plot of the log-transformed values shows a nice bell curve (Gaussian distribution) and the corresponding ECDF shows a nice sigmoidal shape (Figure 8-5).



- Figure 8-5. Distribution of the logarithm of the number of inhabitants in US counties.

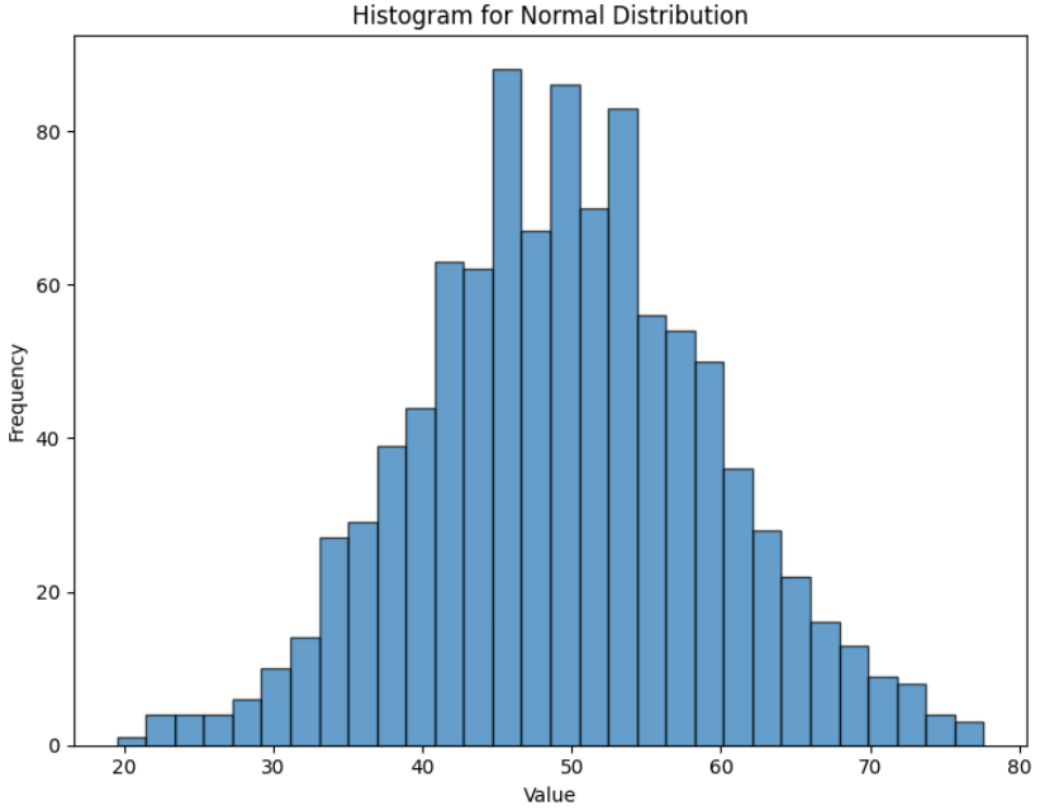
(a) Density plot. (b) Empirical cumulative distribution function. Data source: 2010 US Decennial Census.

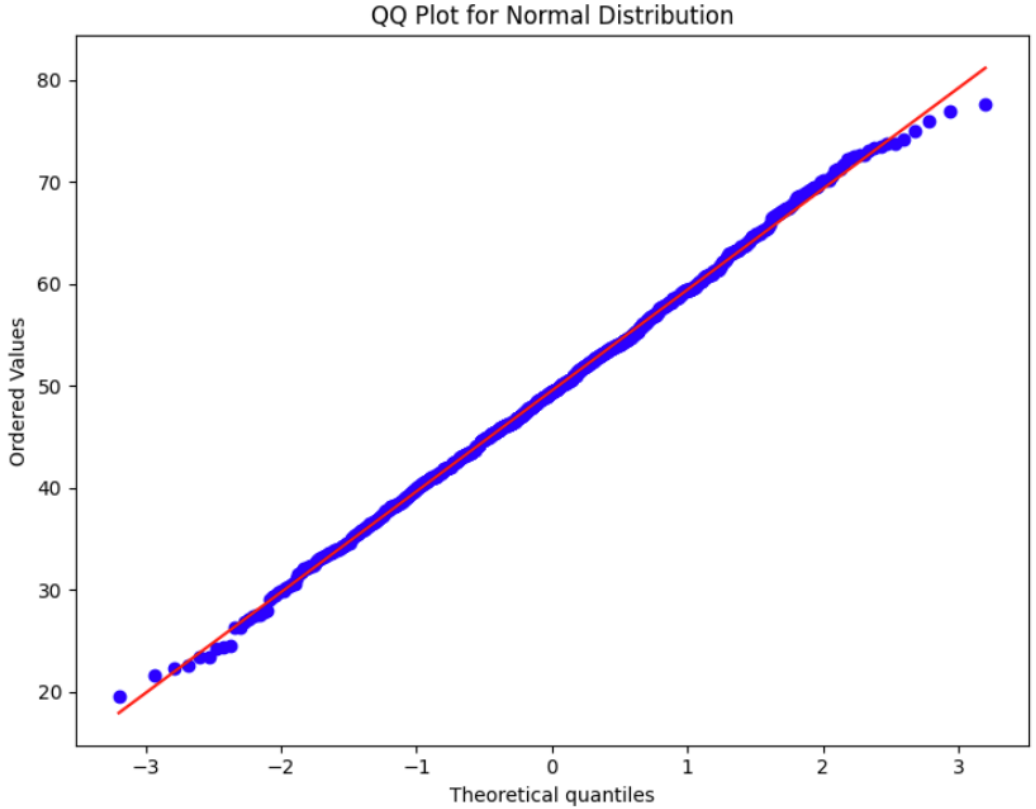
Quantile-Quantile Plot

- Determine if a dataset follows a particular theoretical distribution.
- q-q plots provide a visual comparison between the quantiles of the observed data and the quantiles of a specified theoretical distribution, usually the **standard normal distribution**.
- q-q plots based on ranking the data and visualizing the relationship between ranks and actual values.

Understanding Quantile-Quantile Plot

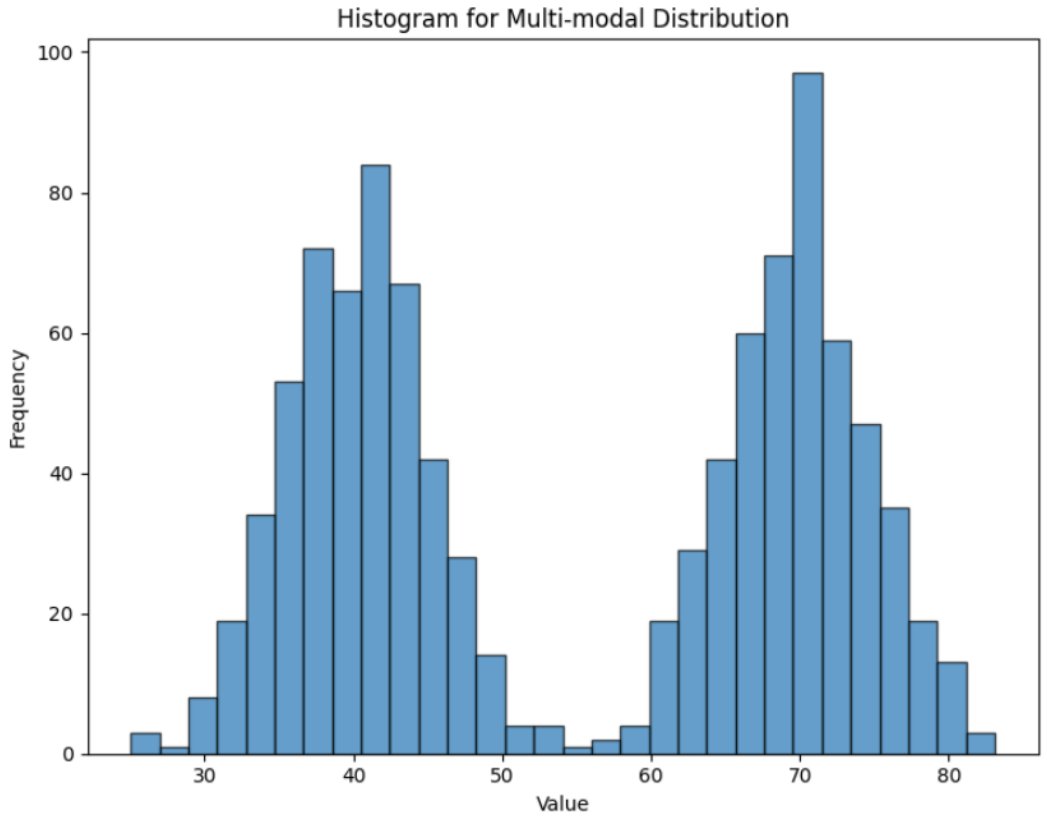
- Theoretical Quantiles (x-axis): Quantiles from the assumed reference distribution.
- Observed Quantiles (y-axis): Quantiles from the observed dataset.
- A **straight line** in a q-q plot indicates the data follows the reference distribution

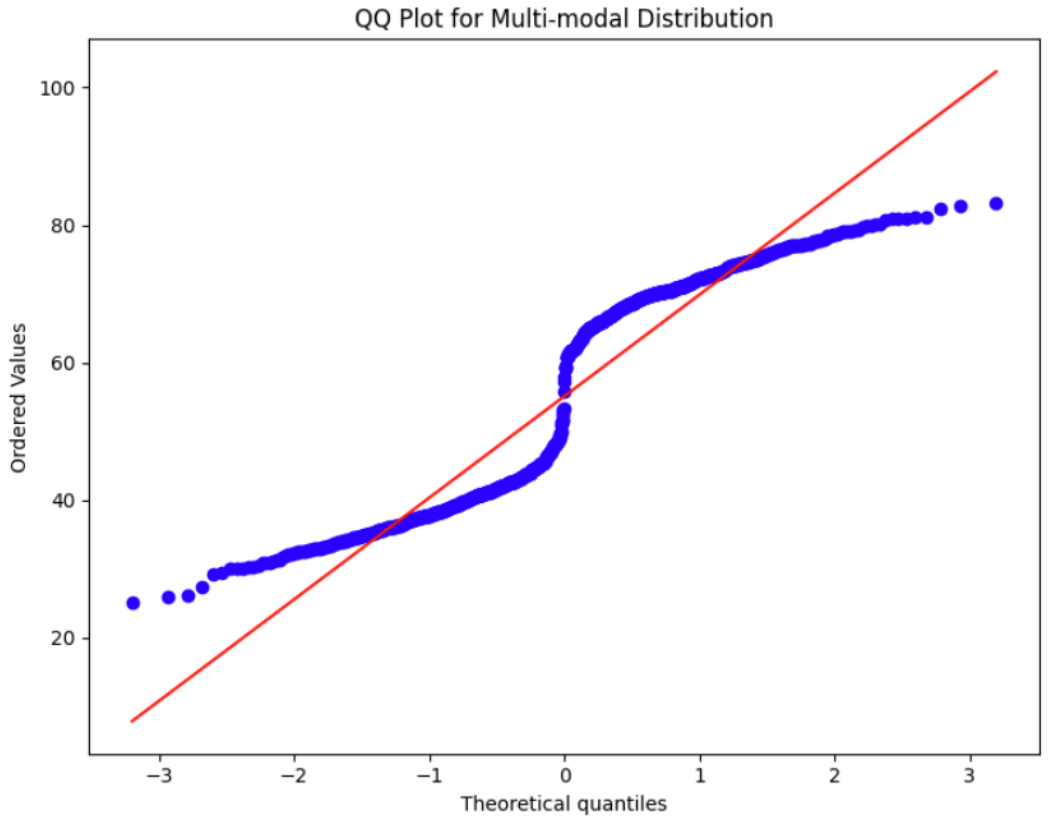




Understanding Quantile-Quantile Plot

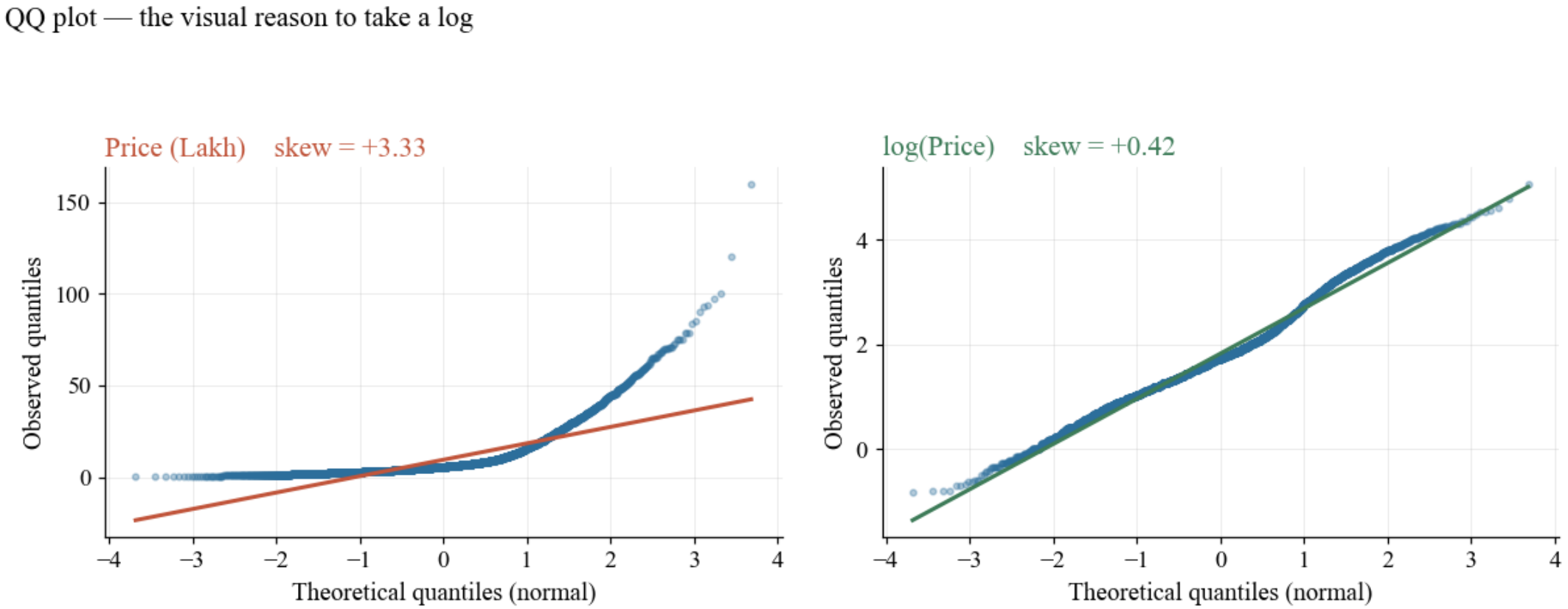
- Theoretical Quantiles (x-axis): Quantiles from the assumed reference distribution.
- Observed Quantiles (y-axis): Quantiles from the observed dataset.
- A **straight line** in a q-q plot indicates the data follows the reference distribution.





QQ Plot: the Visual Reason to Take a Log

- **Your quantiles against a normal distribution's. On the line means normal.**



Price skew **+3.33** → log(Price) skew **+0.42**. A before and after you can show.

TODAY'S LECTURE

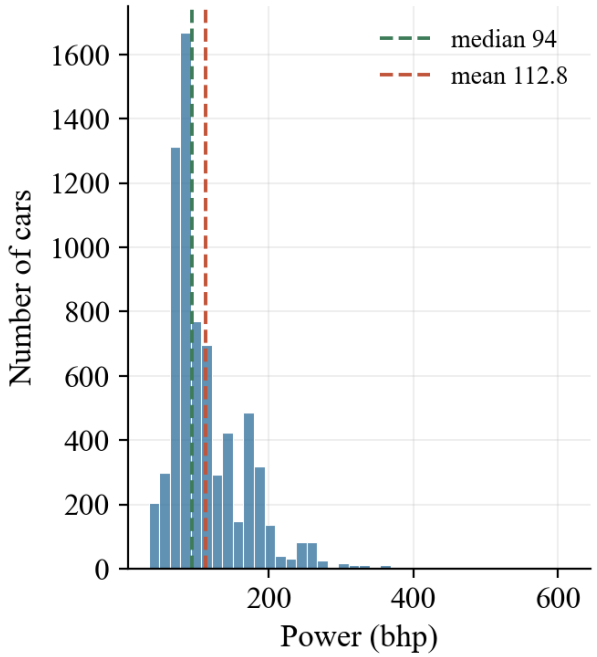
- Review Last Session
- **EDA Task**
- Uncertainty Plot
- Visualizing Multiple Distributions

Univariate Numerics: Power

Power — n = 7,077, 175 missing

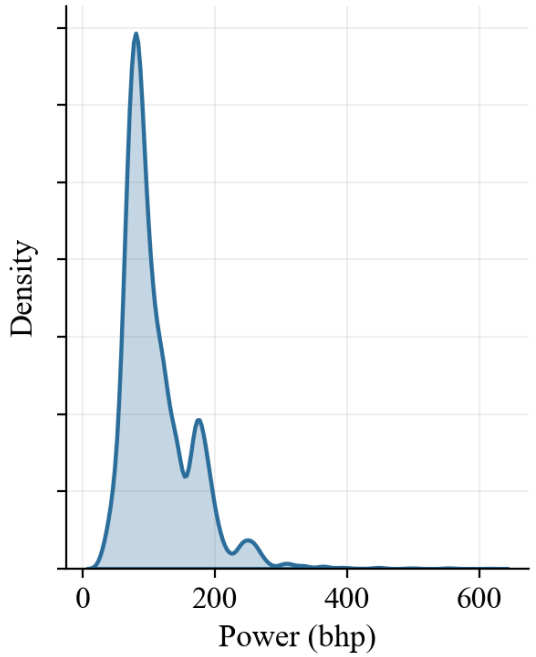
1 · Histogram

the mean sits 20% above the median



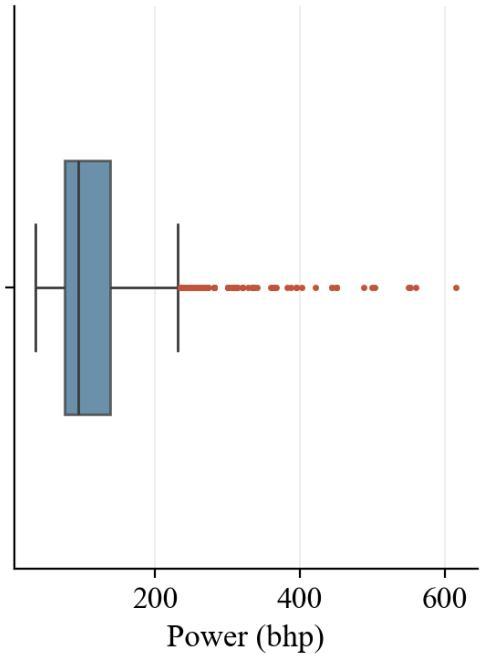
2 · Density

skew +1.96 (right tail)



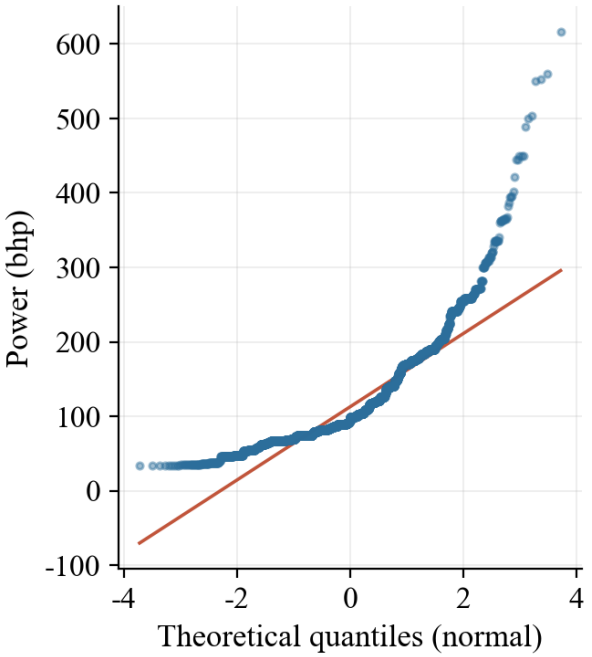
3 · Box plot

273 rows past the fences (3.9%)



4 · QQ plot

R^2 = 0.842 (bends away)



A long right tail of performance cars. The mean sits 20% above the median.

Target or Predictor: Two Different Questions

Same skew, two different jobs. **Only one of these columns is the thing being predicted.**

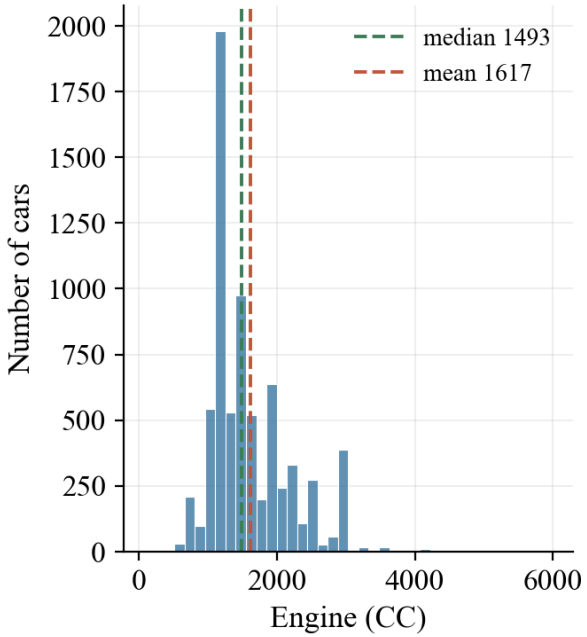
The question	Price — the target	Power — a predictor
What its shape decides	the shape of the model's errors	nothing: a regression assumes nothing about the distribution of an input
Skew, and the range it spans	+3.33 over 0.44 → 160 Lakh (364x)	+1.92 over 34 → 560 bhp (16x)
Residual spread, cheap third → dear third	1.80 → 11.69 Lakh raw (6.5x); 0.46 → 0.65 logged (1.4x)	a predictor has no residuals — the question does not arise
What the log buys	one model equally wrong in percent at every price	R^2 0.591 → 0.626
What the log costs	predictions come back in logs and have to be transformed back	"+1 bhp adds x%" becomes an elasticity
What is left after it	a single smooth right tail — the log fixes it	three bumps, real market segments — a spline or a segment flag fixes those, a log does not

Univariate Numerics: Engine

Engine — n = 7,206, 46 missing

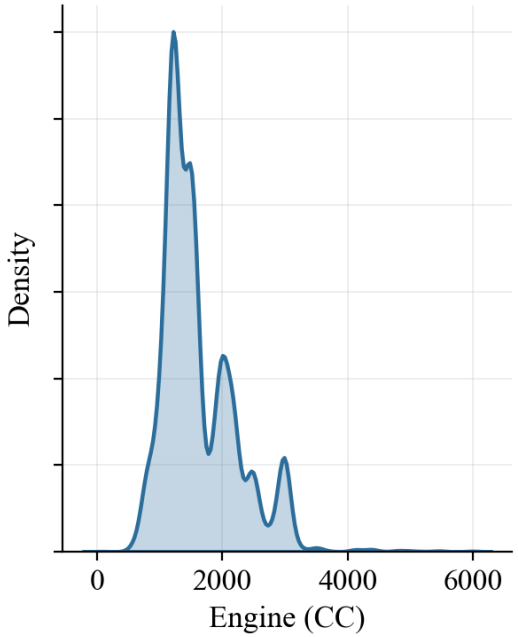
1 · Histogram

the mean sits 8% above the median



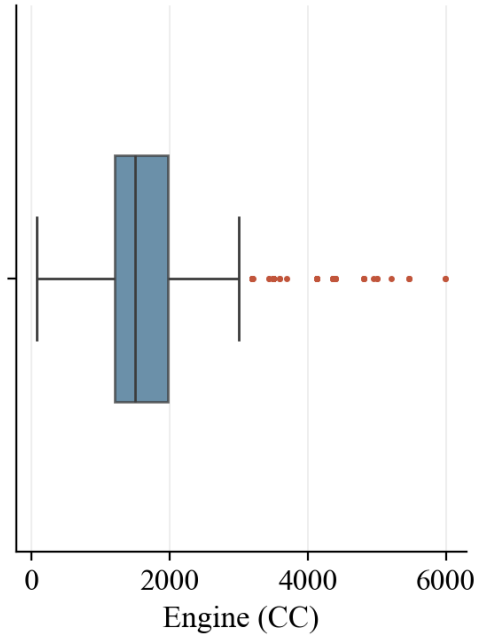
2 · Density

skew +1.41 (right tail)



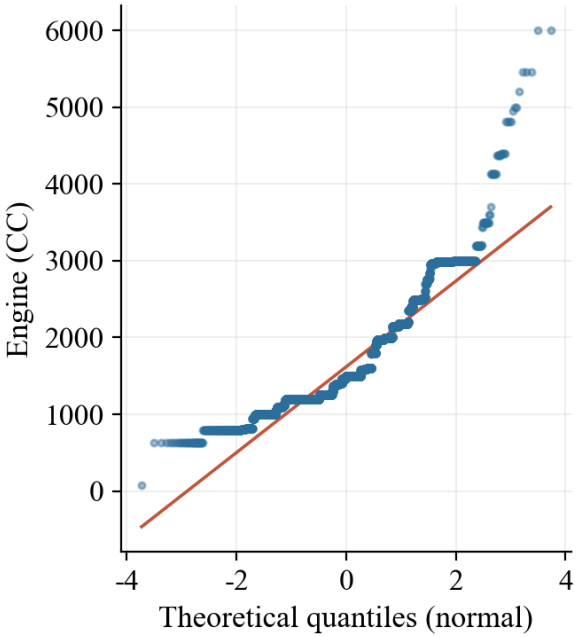
3 · Box plot

65 rows past the fences (0.9%)



4 · QQ plot

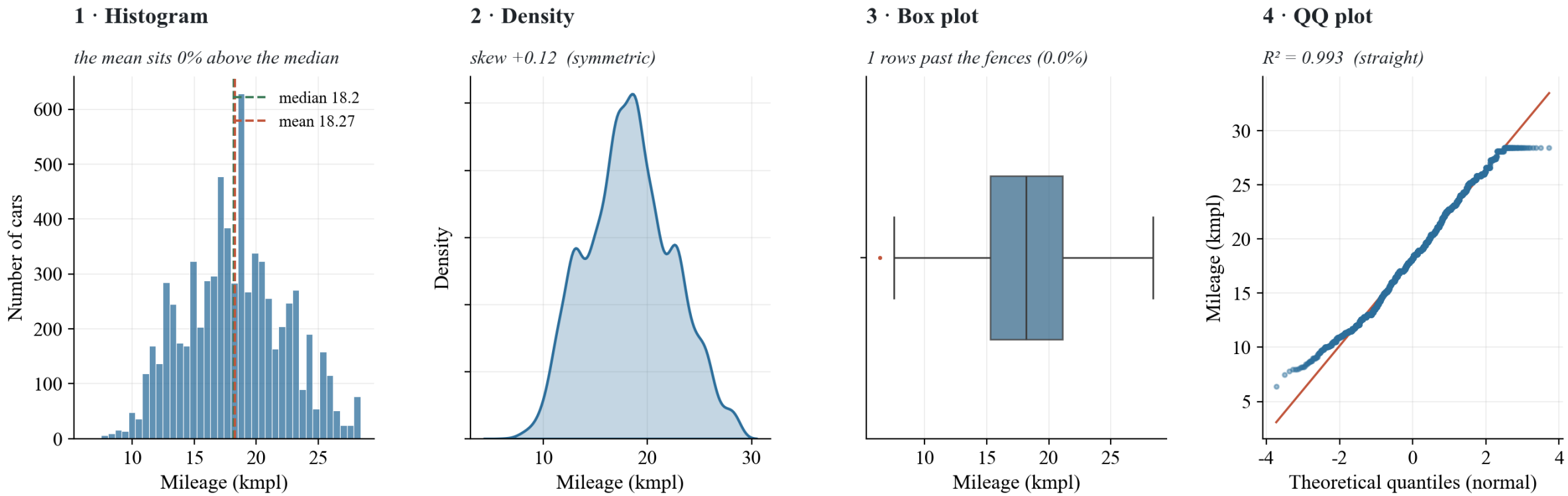
R^2 = 0.880 (bends away)



The histogram has teeth; the density smooths them away. Engines are sold in standard sizes, 1197, 1248, 1498 CC, so this column behaves more like ordered categories than like a measurement.

Univariate Numerics: Mileage

Mileage — n = 7,169, 83 missing



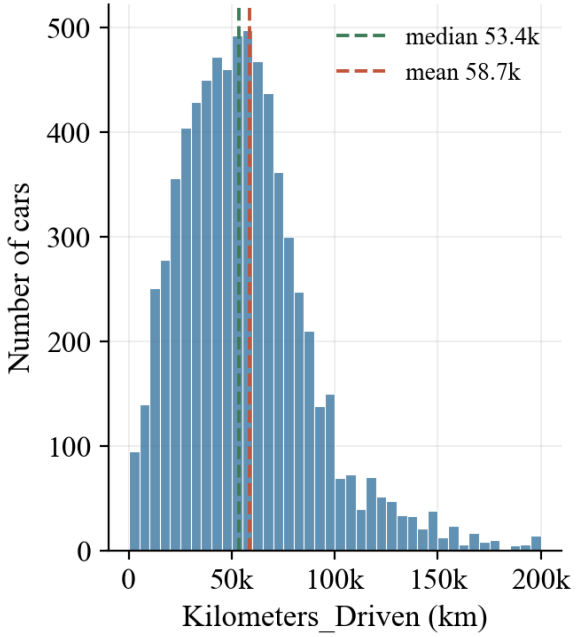
The one near-symmetric column in the file. Skew +0.12, mean and median on top of each other, and a QQ plot that stays on the line, the only column here where the mean is safe to use.

Univariate Numerics: Kilometers_Driven

Kilometers_Driven — n = 7,252, 0 missing (axis clipped at the 99.5th pct: 34 rows above, max 6,500,000)

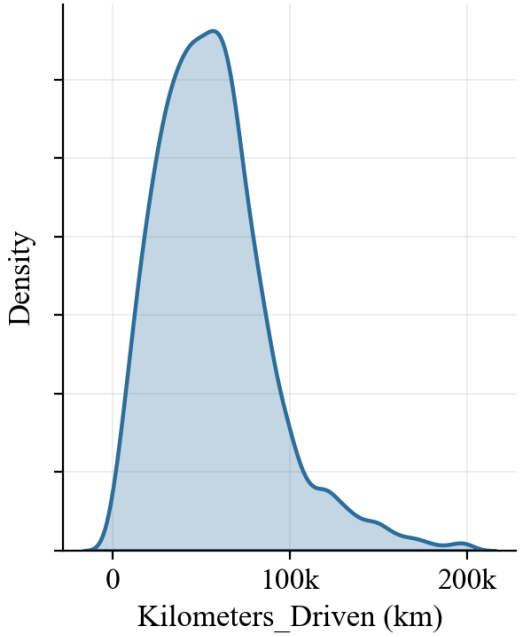
1 · Histogram

the mean sits 10% above the median



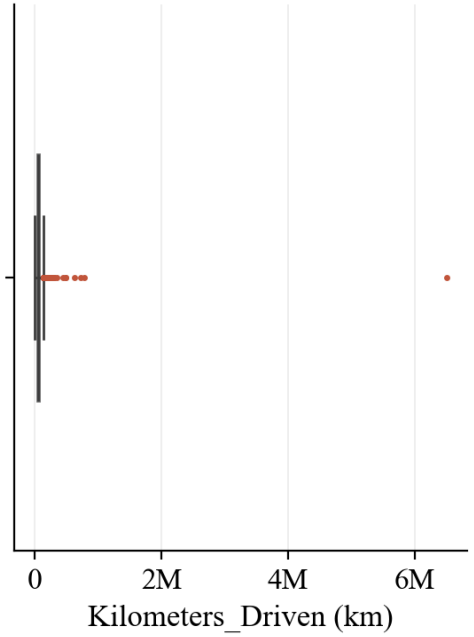
2 · Density

skew +61.57 (right tail)



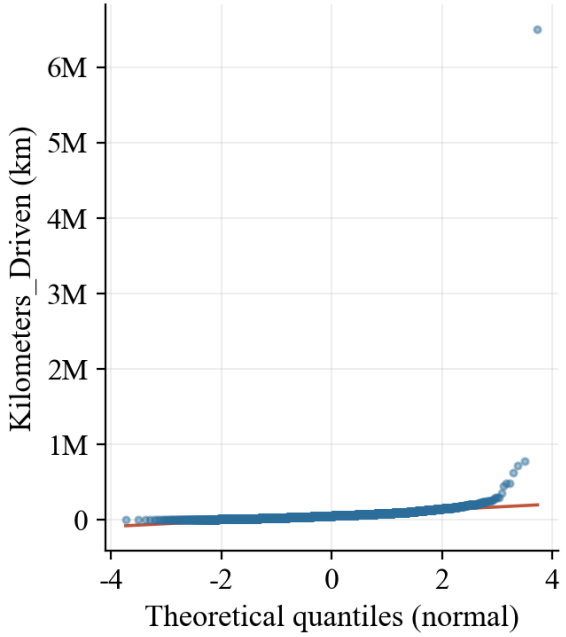
3 · Box plot

258 rows past the fences (3.6%)



4 · QQ plot

R^2 = 0.192 (bends away)



Name	Year	Kilometers_Driven	Price	Transmission
BMW X5 xDrive 30d M Sport	2017	6500000	65.0	Automatic
Skoda Octavia Ambition Plus 2.0 TDI AT	2013	775000	7.5	Automatic
Volkswagen Vento Diesel Highline	2013	720000	5.9	Manual

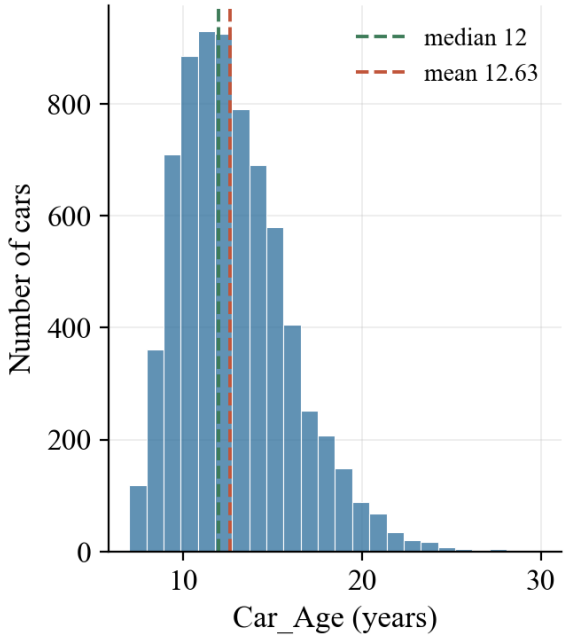
One row reads 6,500,000 km.

Univariate Numerics: Car_Age

Car_Age — n = 7,252, 0 missing

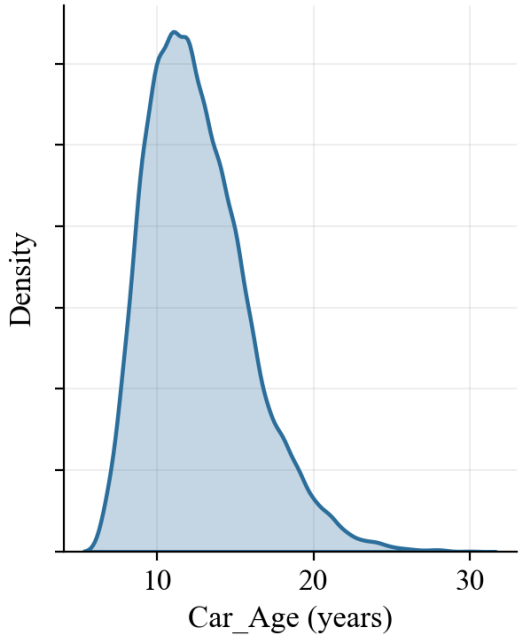
1 · Histogram

the mean sits 5% above the median



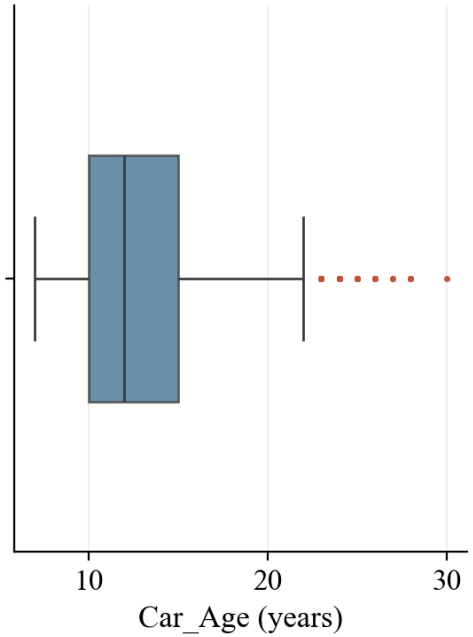
2 · Density

skew +0.84 (right tail)



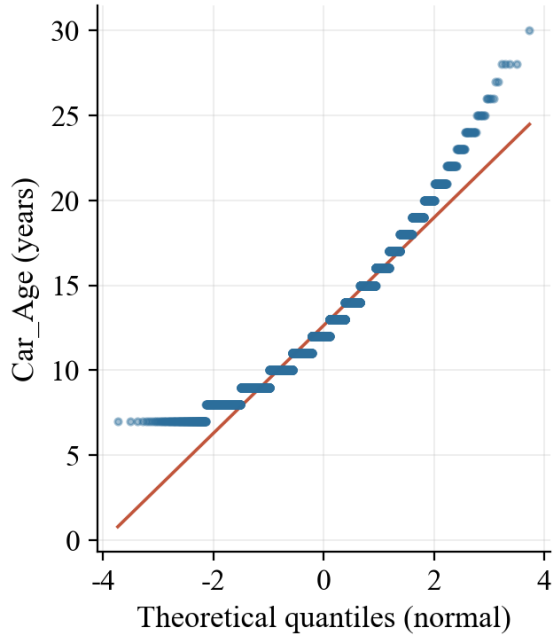
3 · Box plot

58 rows past the fences (0.8%)



4 · QQ plot

$R^2 = 0.951$ (bends away)

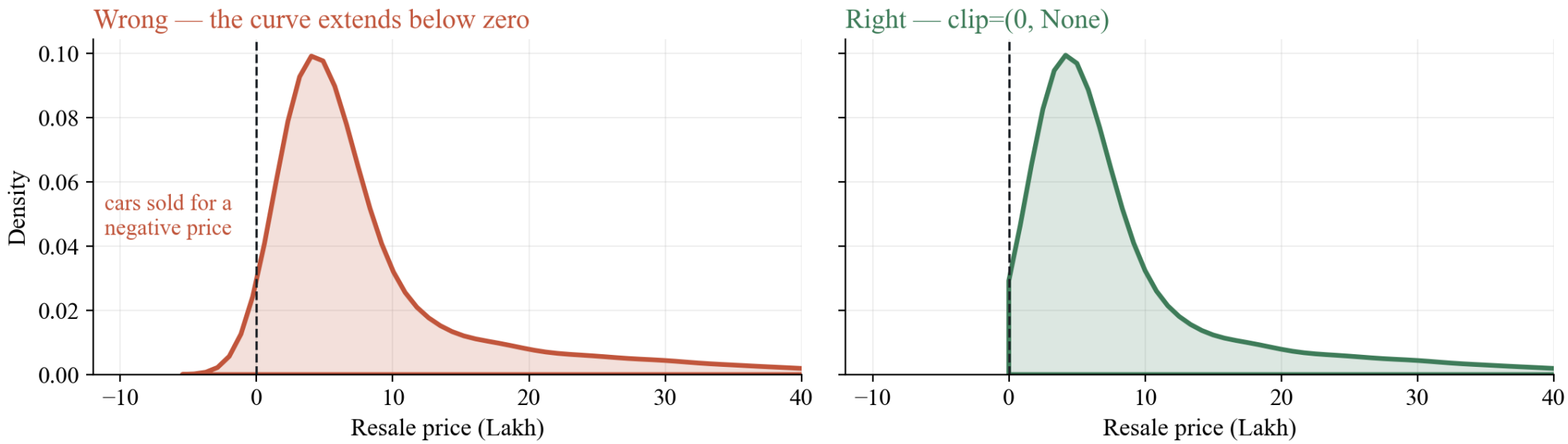


Read the histogram here and ignore the curve. Car_Age is an integer count from 7 to 30 years, and a density plot draws a smooth line over a count — convenient, and slightly a lie.

The Wrong Figure: a KDE Invents Negative Prices

- **Nothing stops the curve running past a hard boundary.**

A KDE will happily produce data that cannot exist



sns.kdeplot(price, **clip=(0, None)**) one argument, and the figure stops lying.

The Output of an EDA Is a Decision Table

Structure – Clean – Univariate – Bivariate – Multivariate – Decisions

Column / issue	Decision	Because
Mileage, Engine, Power, Seats	impute with the median	Most of them skewed; the mean follows the tail
Price	drop rows where missing	it is the target; never impute a target
Price, for modelling	use log(Price)	skew **3.33 ->** 0.42
The 6,500,000 km row	flag, drop it	entry error, not an extreme car

Act 3, Closed

- **Five tools, five different questions**
  - Histogram — the shape, if you are honest about bin width.
  - Density — the same shape, smoothed; watch the boundaries.
  - Box plot — five numbers, fast comparison, hides everything else.
  - Violin — the shape a box plot hides. Use it whenever groups might be mixed.
  - QQ plot — is this normal, and did my transform help?

Bivariate Analysis

Structure – Clean – Univariate – **Bivariate** – Multivariate – Decisions

➢ ACT 4

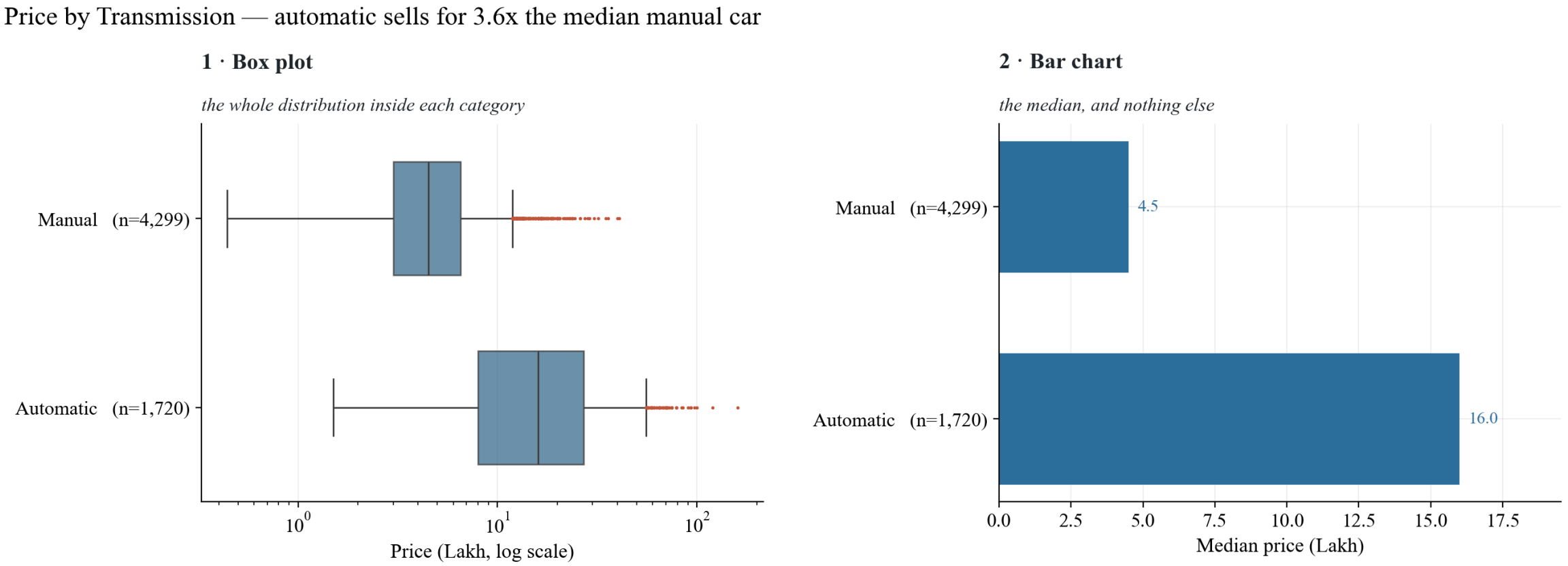
- What moves with price?
- Three kinds of pair, and a different set of charts for each.

Three Kinds of Pair

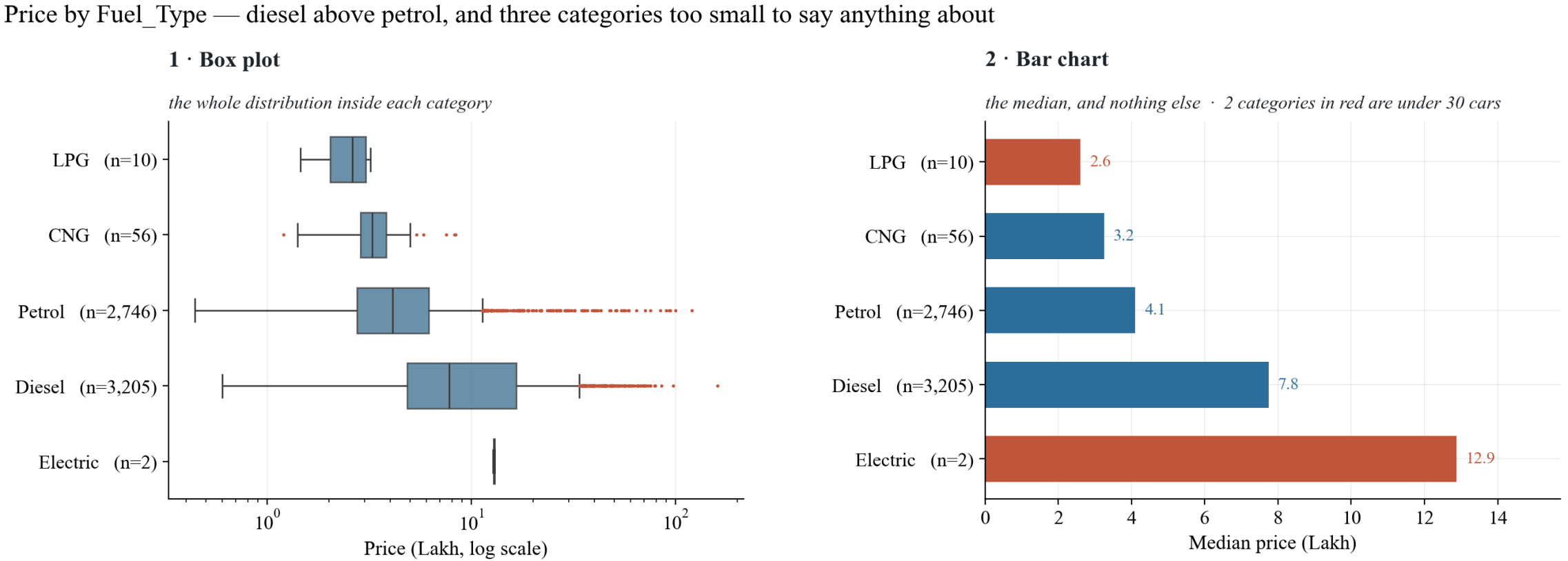
Pair	Charts	Statistics	Tonight's example
numeric x numeric	scatter plot	Pearson, Spearman	Price vs Power, Price vs Kilometers_Driven
numeric x categorical	box, violin, faceted density	t-test, ANOVA	Price by Transmission, Mileage by Fuel_Type
categorical x categorical	grouped bar, 100% stacked bar, heatmap	chi-square	Fuel_Type x Transmission

Price by Transmission

One numeric column, one categorical column. The box shows every car; the bar shows one number per category. Price axis is log.



Price by Fuel_Type



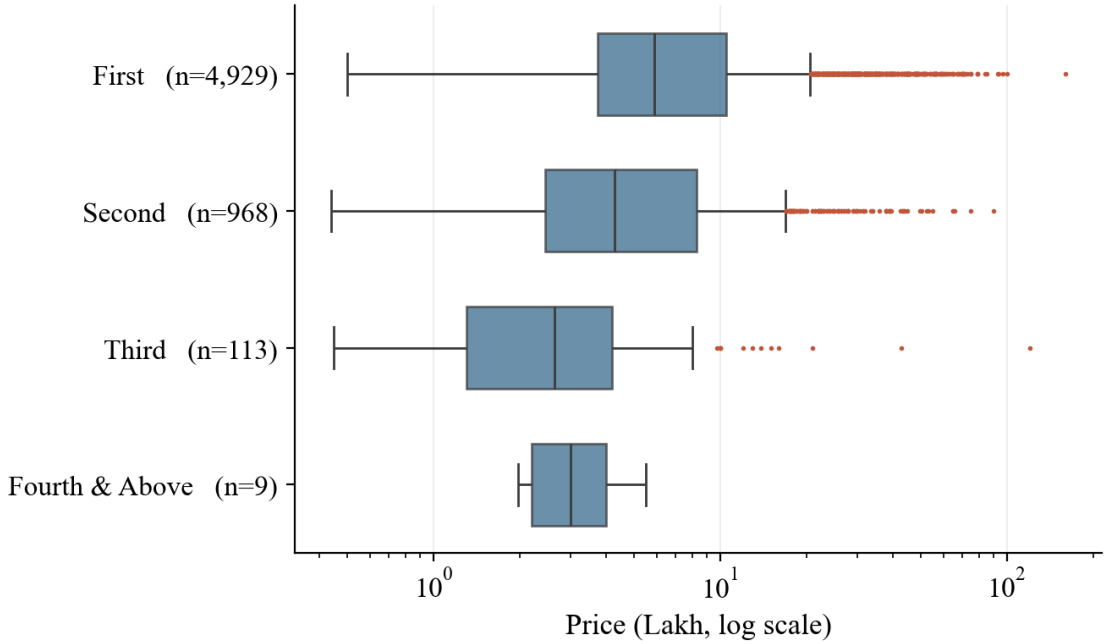
The bar chart draws Electric (n=2) exactly as confidently as Diesel (n=3,205). The box plot shows one of them is a line, not a distribution. Count before you compare.

Price by Owner_Type

Price by Owner_Type — each extra owner costs money, until the counts run out

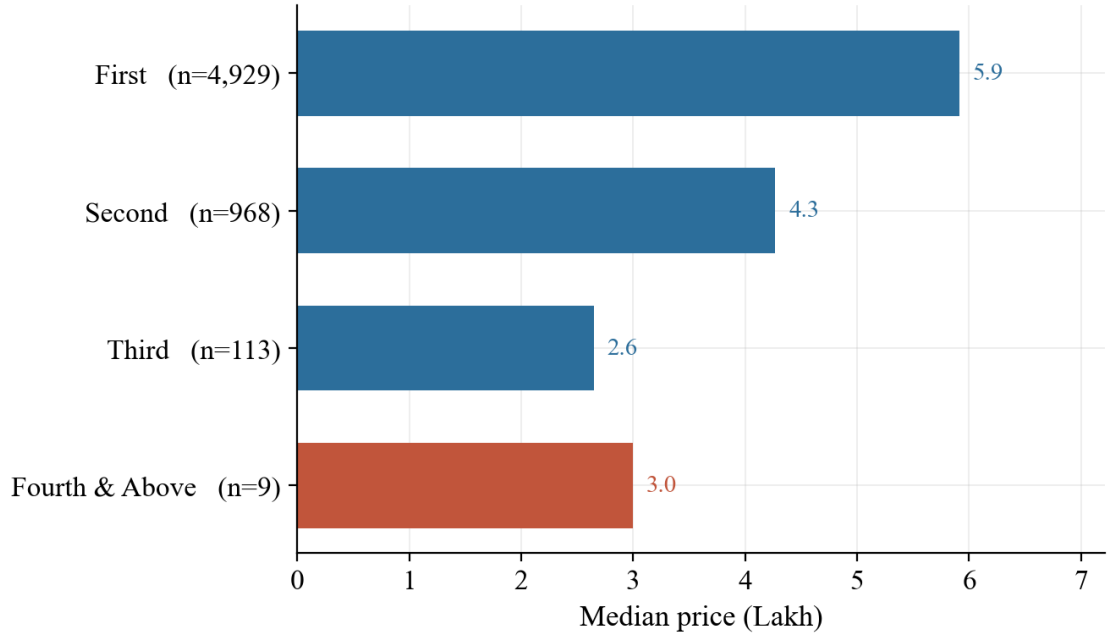
1 · Box plot

the whole distribution inside each category



2 · Bar chart

the median, and nothing else · 1 category in red is under 30 cars



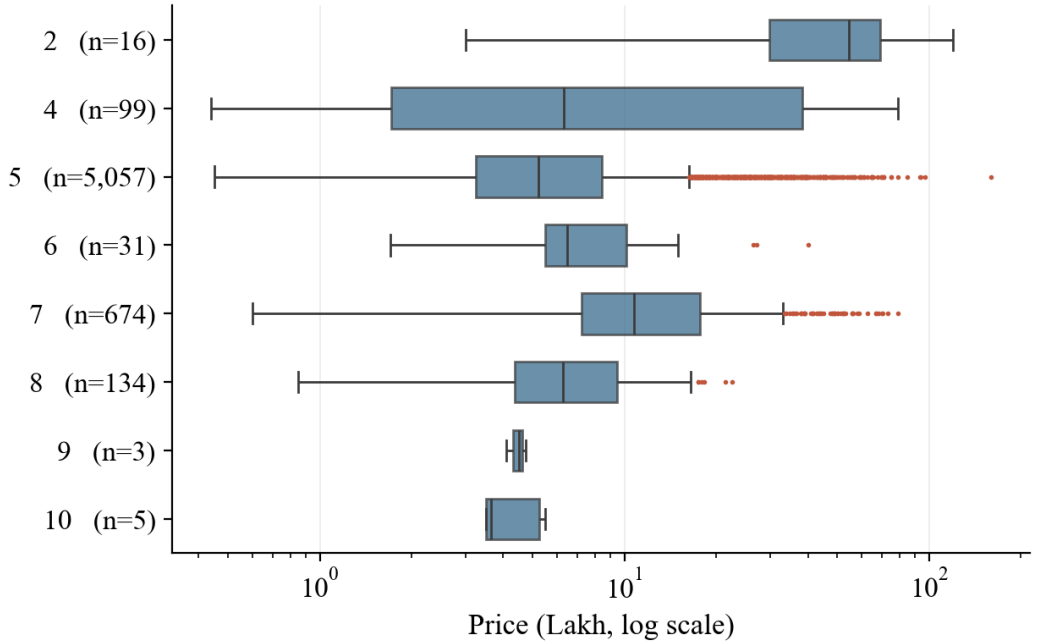
Fourth & Above on nine cars. Read the count before you read the pattern.

Price by Seats

Price by Seats — the two-seaters are the most expensive cars in the file, and there are 16 of them

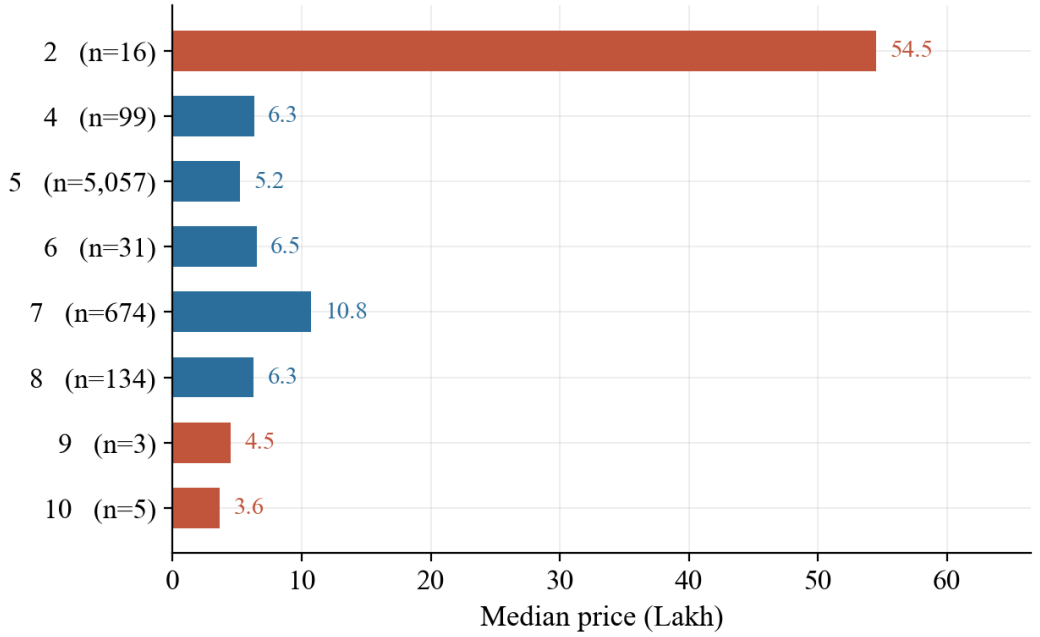
1 · Box plot

the whole distribution inside each category



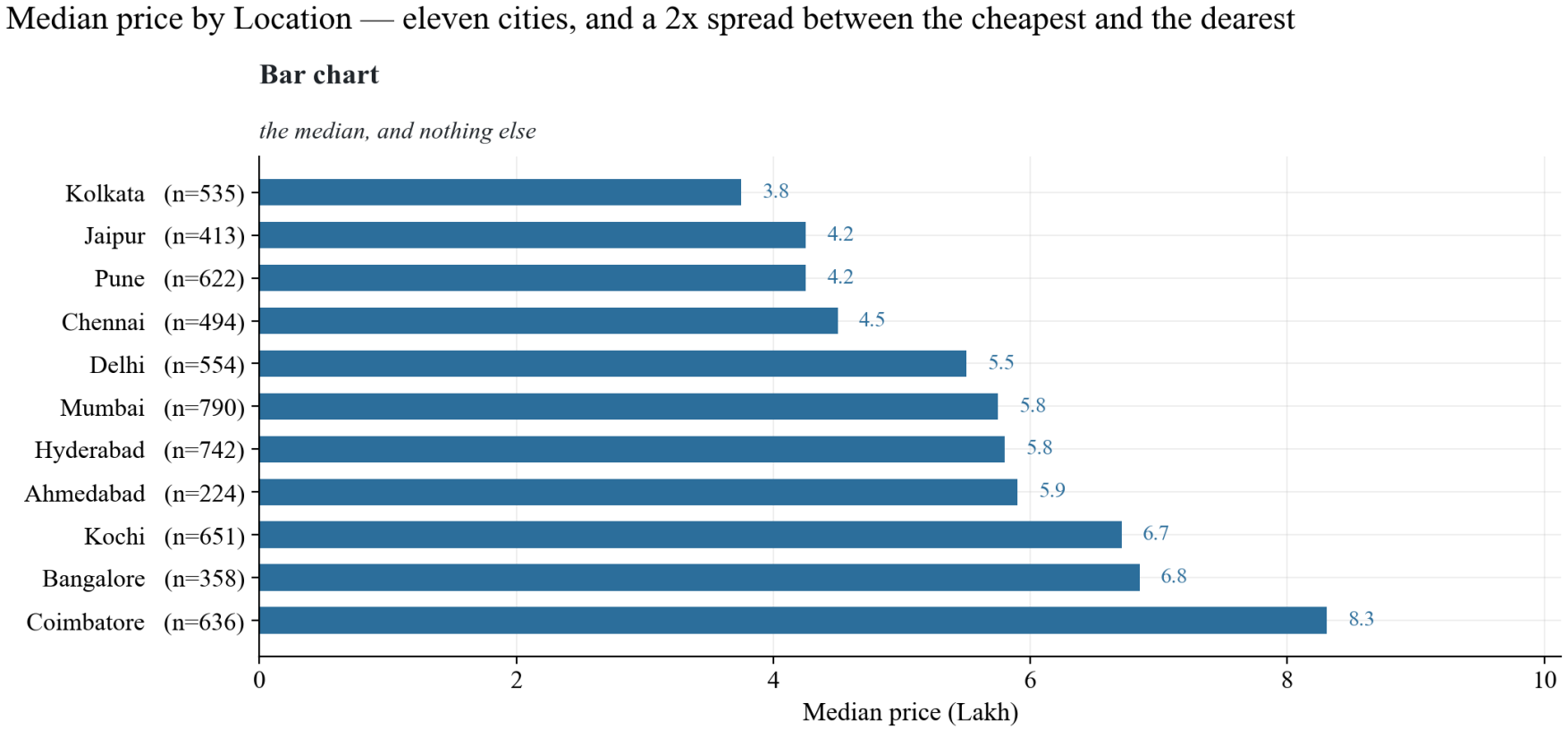
2 · Bar chart

the median, and nothing else · 3 categories in red are under 30 cars



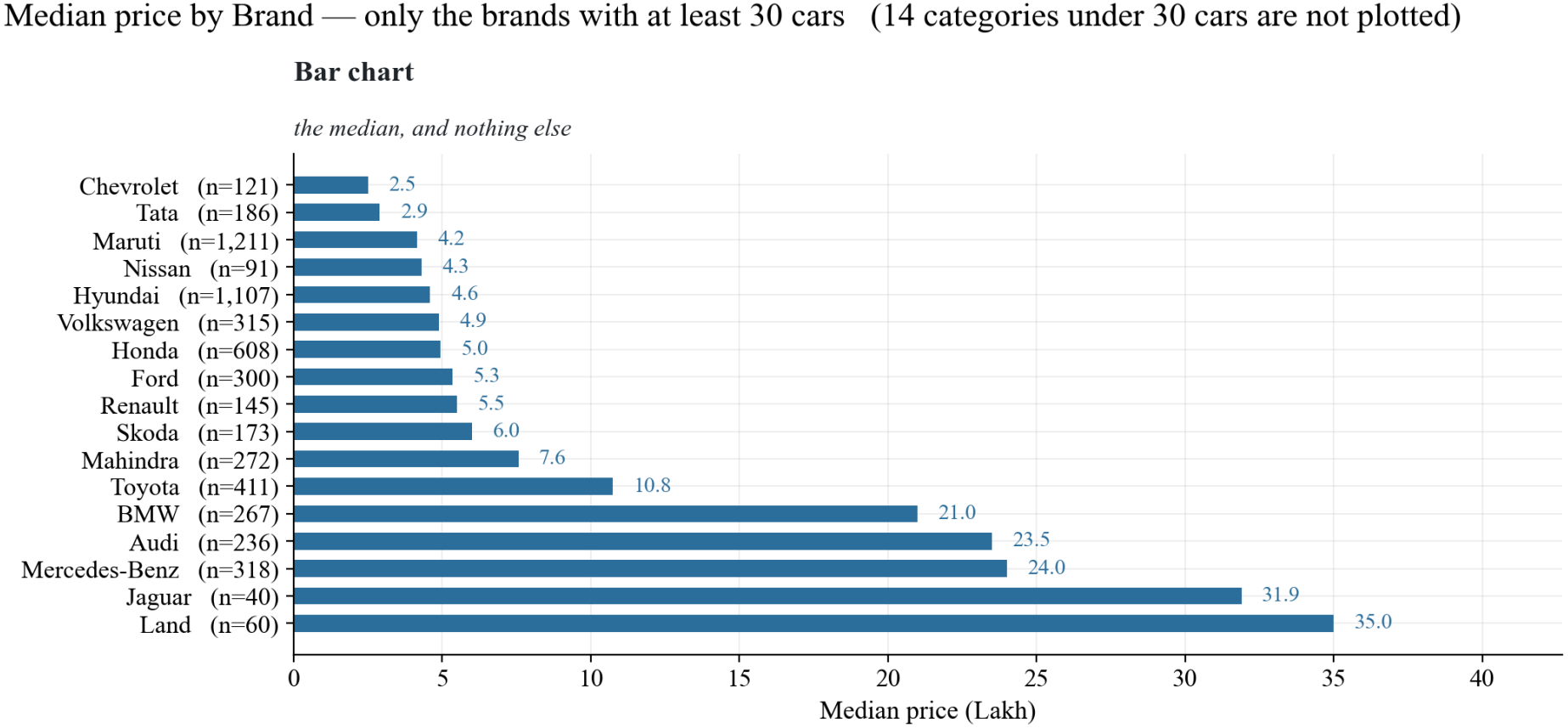
Two-seaters have the highest median price in the whole file, because a two-seater is a sports car. And the box plot adds what the bar hides: that group is also the widest, so the median is standing in for very little.

Median Price by Location



Eleven cities and a 2.2x spread, from Kolkata at 3.8 to Coimbatore at 8.3. No box plots here: past about a dozen categories a row of boxes stops being readable, and the bar chart is what survives.

Median Price by Brand



The widest spread of any column: about 14x from Chevrolet to Land Rover. Fourteen brands with under 30 cars are not plotted at all.

Break: 10 minutes

Multivariate Analysis

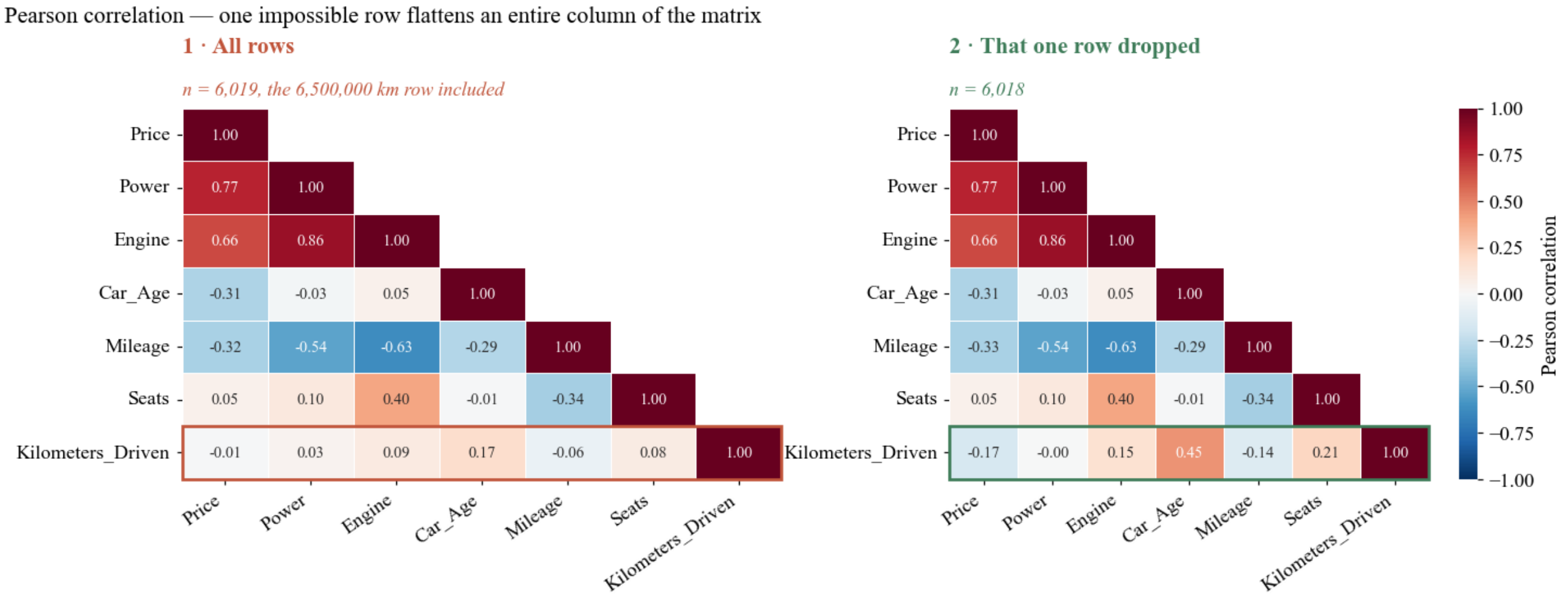
Structure – Clean – Univariate – Bivariate – Multivariate – Decisions

➢ ACT 5

- **What is really driving price, once you hold something else constant?**
- The last two slides of this act reverse a conclusion we have already drawn.

The Correlation Heatmap

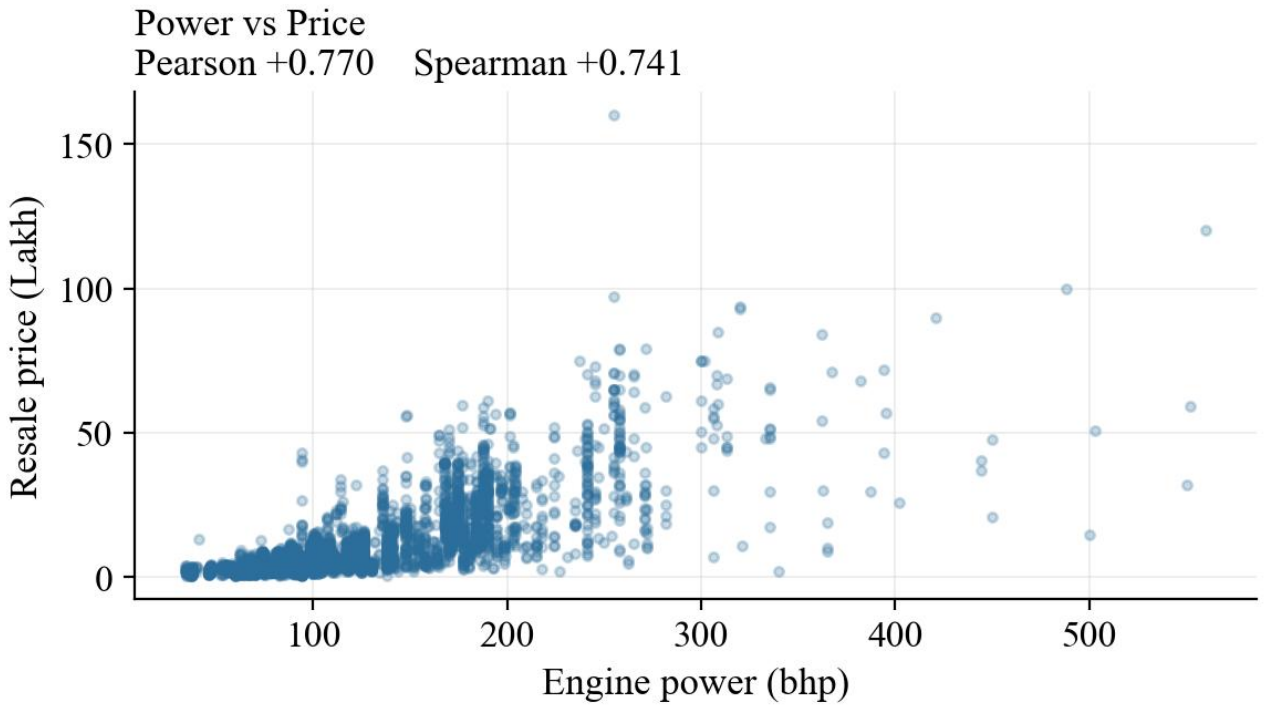
- **Diverging palette, zero at the center, the Week 3 rule.**

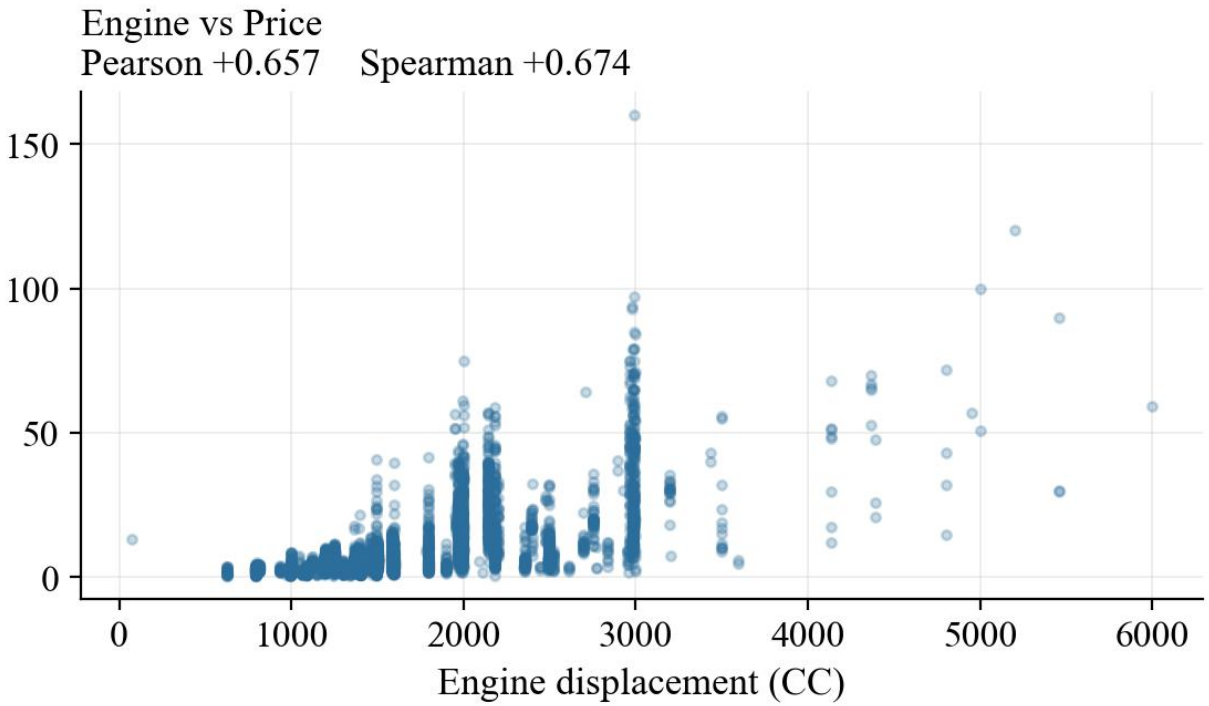


The Two Strongest Predictors

- **Power and Engine. Pearson and Spearman agree, which means the relationship is roughly linear.**

The two strongest predictors in the file





The Output of an EDA Is a Decision Table

Structure – Clean – Univariate – Bivariate – Multivariate – Decisions

Column / issue	Decision	Because
S.No.	drop the column	row number; it also disabled duplicated()
Mileage, Engine, Power	strip units, cast to float	describe() was silently skipping them
New_Price/Mileage unit convert	convert Cr to Lakh (x100)	1 Cr = 100 Lakh; 20 rows
Power == 'null bhp'	treat as missing	129 rows; true missing rate 0.6% -> 2.4%
Mileage == 0 / Seats == 0	treat as missing	81 rows and 1 row; both impossible
duplicate row	drop	only visible once S.No. was removed
New_Price	drop the column	86% missing
Price	drop rows where missing	it is the target; never impute a target

The Output of an EDA Is a Decision Table

Structure – Clean – Univariate – Bivariate – Multivariate – Decisions

Column / issue	Decision	Because
Mileage, Engine, Power, Seats	impute with the median	Most skewed; the mean follows the tail
Price, for modelling	use log(Price)	skew **3.33 ->** 0.42
The 6,500,000 km row	flag, drop it	entry error, not an extreme car

Six Traps, and Not One of Them Raised an Error

#	The trap	What it cost
1	describe() silently drops 3 columns	the three strongest numeric predictors, invisible
2	'null bhp' hides inside a string	the missing rate wrong by a factor of 4
3	0 kmpl and 0 seats are not NaN	82 impossible values inside every mean
4	a 2017 BMW with 6,500,000 km	kurtosis of 4,670 and a broken correlation
5	one row flattens Pearson to -0.011	a real feature discarded as useless

Every one of these runs without an error message. You find them by drawing the figure, or by knowing what the column means.

What Was the EDA Worth?

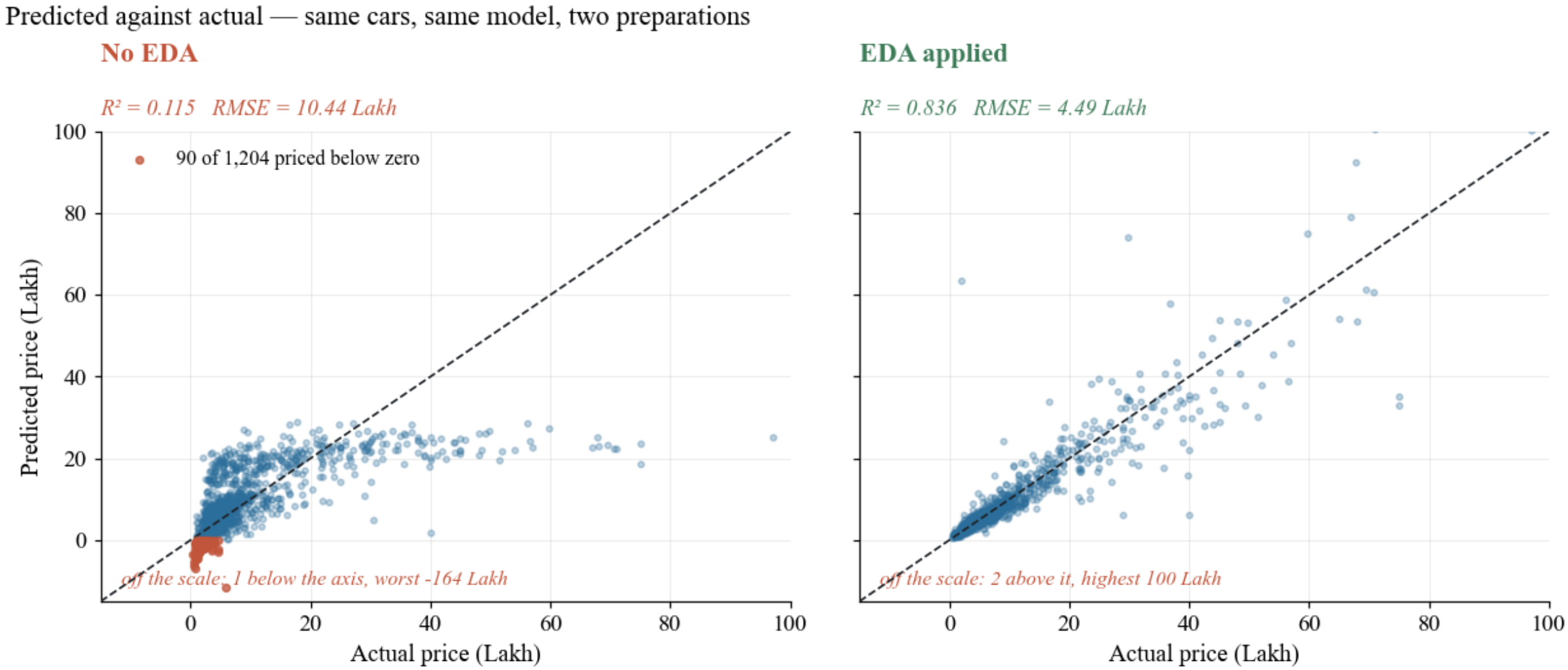
	Path A — no EDA	Path B — the decision table applied
preparation	`read_csv`, keep what is already numeric, one-hot the four clean categorical columns, fill gaps with the mean	all eleven decisions from slides 39–40, plus `Car_Age` and `Brand`
target	`Price`	`log(Price)`, back-transformed before scoring
model	`LinearRegression()`	`LinearRegression()`

What Was the EDA Worth?

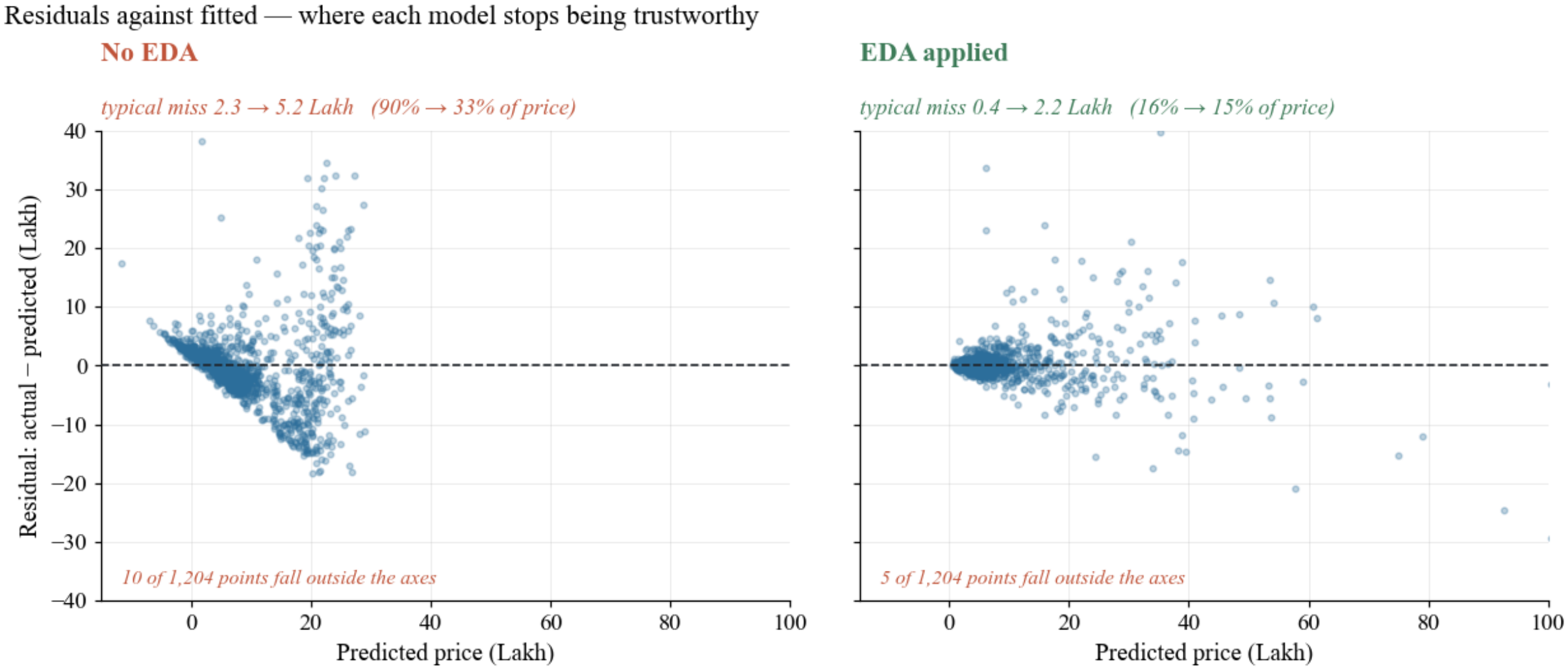
The rules that keep the comparison honest

1. **The same cars.** One 80/20 split, drawn once, used by both paths.
2. **The same test set, scored on the same scale.** Path B predicts logs, so its predictions are exponentiated back to Lakh before anything is measured.
3. **You may clean training data. You may not drop rows from the test set.** The duplicate row and the 6,500,000 km row leave Path B's *training* data; if either lands in the test set it stays there, because in production you have to return a number for every car you are handed.
4. **Every statistic used to fill a gap — mean, median, brand list — is computed on the training rows only,** in both paths. Otherwise the test set leaks.

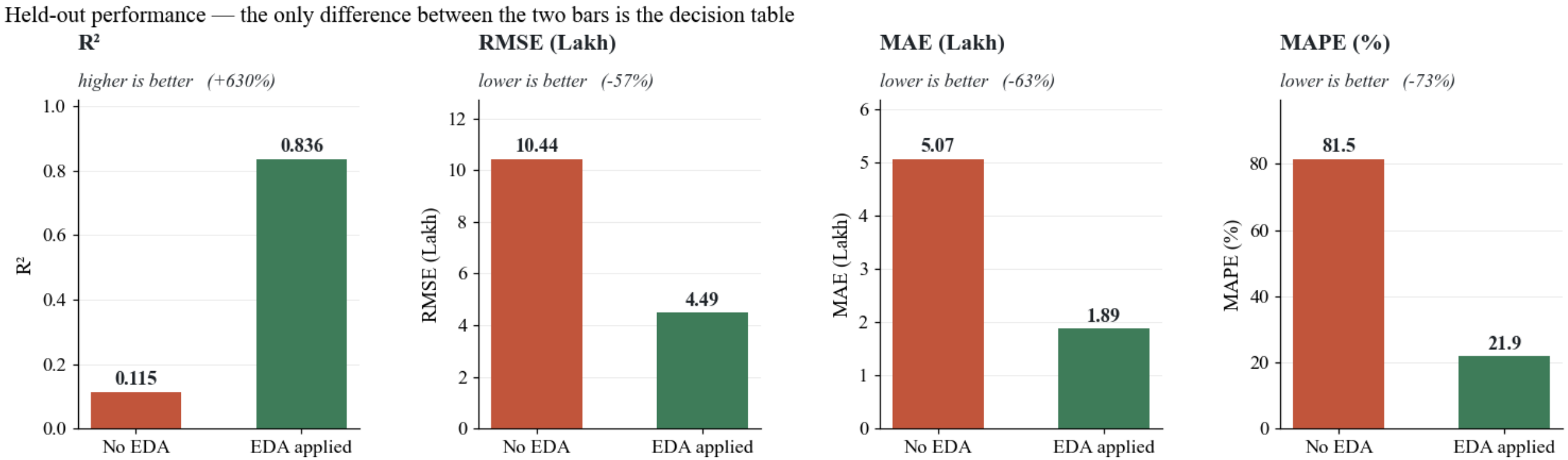
What Was the EDA Worth?



What Was the EDA Worth?



What Was the EDA Worth?



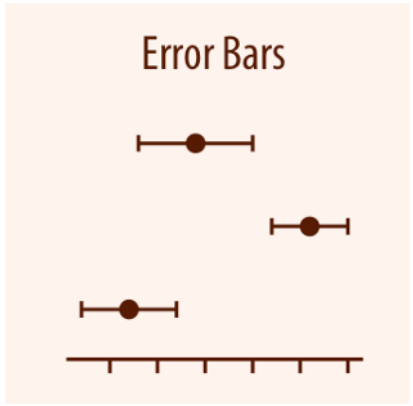
NVIDIA RAPIDS: Accelerate EDA

- Video Demo:
  - https://youtu.be/PJpCJsqcfOk?si=DHF1Q2nsVV2jlh20
- Code Demo:
  - https://colab.research.google.com/gist/will-hill/aa24c3ffe1428c005af3793fcacf9bd2/cudf_pandas_opencellid_demo.ipynb#scrollTo=sIhJowH53A-y
- Accelerate Python module: https://github.com/NVIDIA/accelerated-computing-hub/tree/main/tutorials/accelerated-python
- Source: NVIDIA Developer

TODAY'S LECTURE

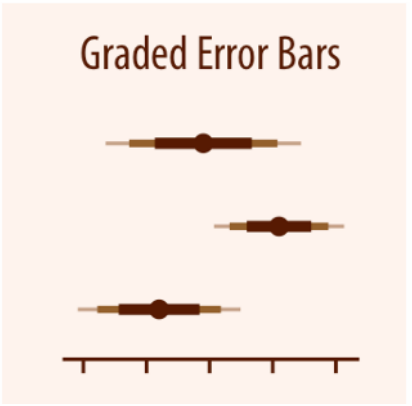
- Review Last Session
- EDA Task
- **Uncertainty Plot**
- Visualizing Multiple Distributions

Directory of Visualizations: Uncertainty

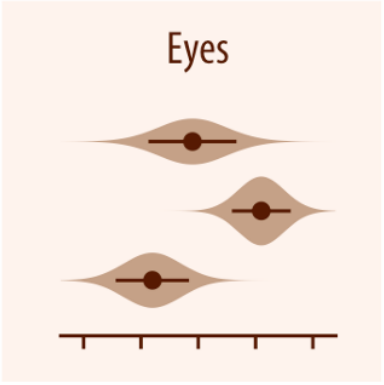




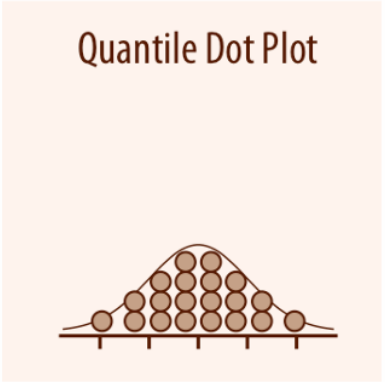




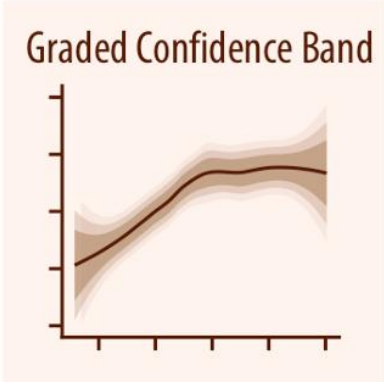


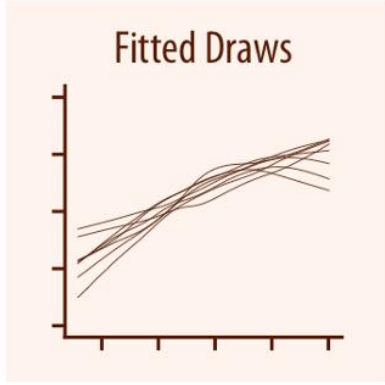








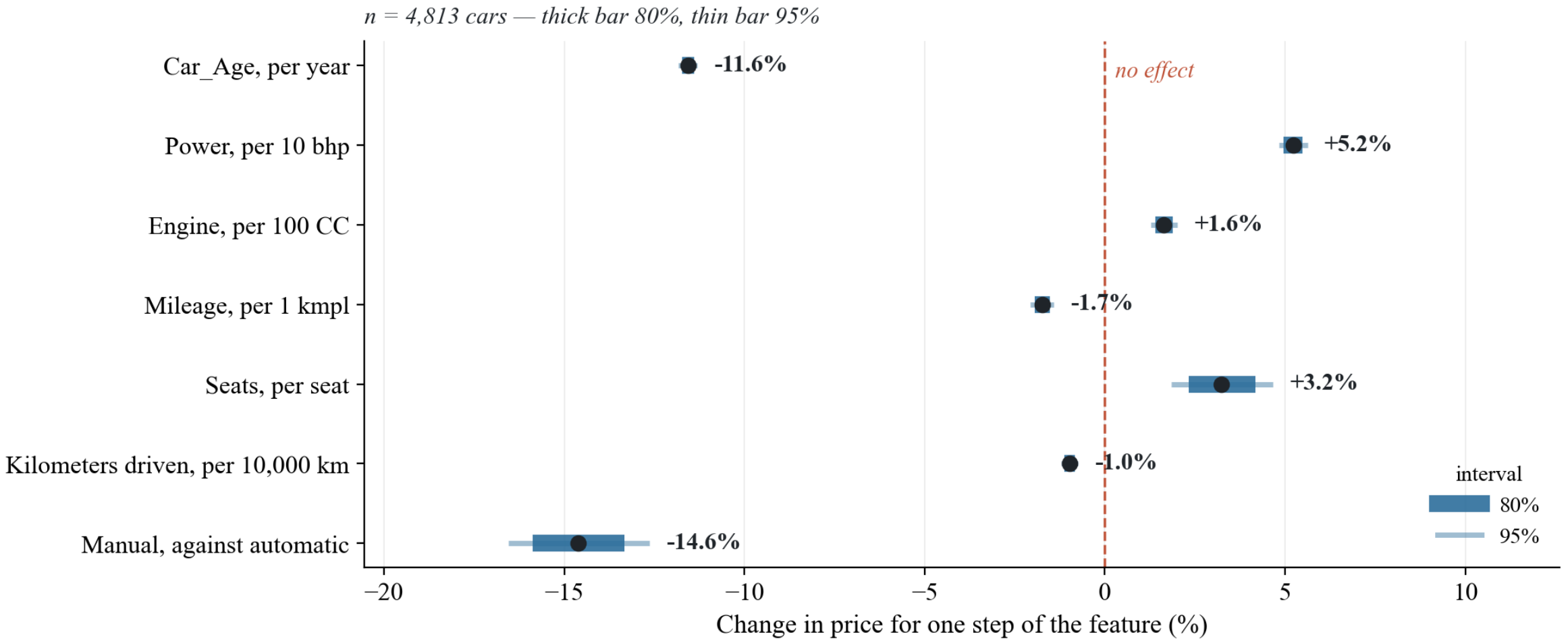




Source: Fundamentals of Data Visualization,
Claus O. Wilke, O'Reilly Media 2019

Uncertainty: How Well Do We Know Each Effect?

What each feature is worth — and how well we know it

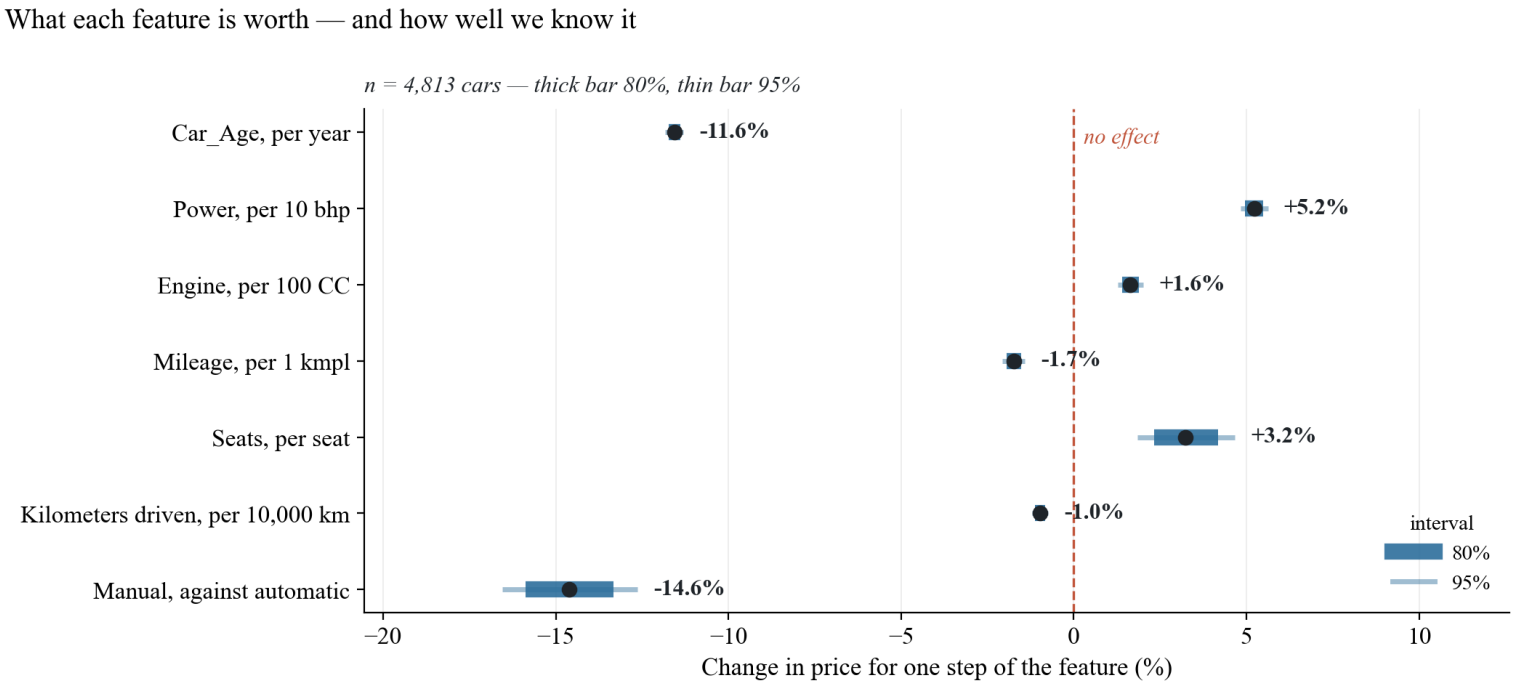


Uncertainty: How Well Do We Know Each Effect?

Feature	Plain-English meaning
Car_Age −11.6%	Each additional year of age, the price drops to 88.4% of its previous value
Power +5.2%	Each extra 10 bhp, the price rises 5.2%
Engine +1.6%	Each extra 100 CC, the price rises 1.6%
Mileage −1.7%	Each extra 1 kmpl of fuel efficiency, the price falls 1.7%
Seats +3.2%	Each extra seat, the price rises 3.2%
Kilometers −1.0%	Each extra 10,000 km driven, the price falls 1.0%
Manual −14.6%	A manual car is 14.6% cheaper than a comparable automatic

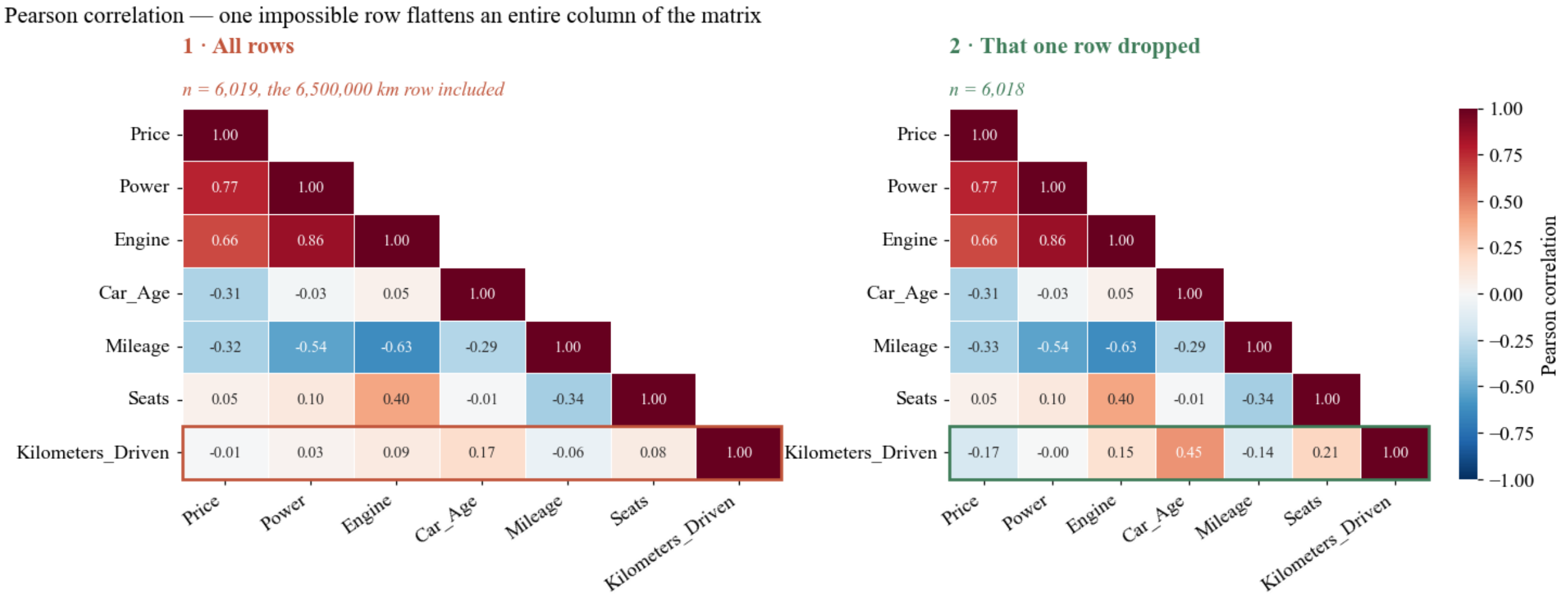
Uncertainty: How Well Do We Know Each Effect?

- Thick bar = 80% confidence interval; thin bar = 95% confidence interval. They show the range of effect sizes consistent with the data.
- Shorter bar = more precise estimate. Car_Age's bar is barely visible (±0.3), so we know the effect is close to −11.6% per year. Seats' bar is wide (+1.8 to +4.7), so the effect is clearly positive, but its exact size is uncertain.
- A bar that doesn't cross the red line is statistically significant. None of the 7 features cross it, so all are significant at that level.



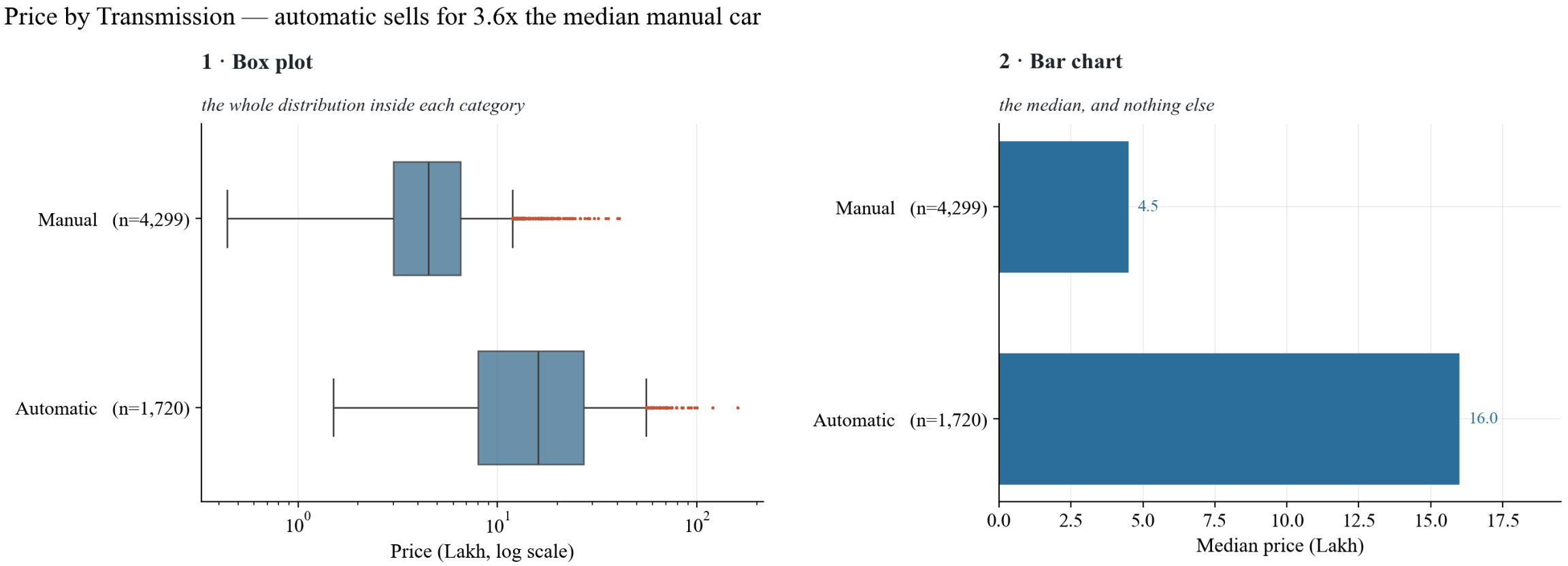
The Correlation Heatmap

- **Diverging palette, zero at the center, the Week 3 rule.**



Price by Transmission

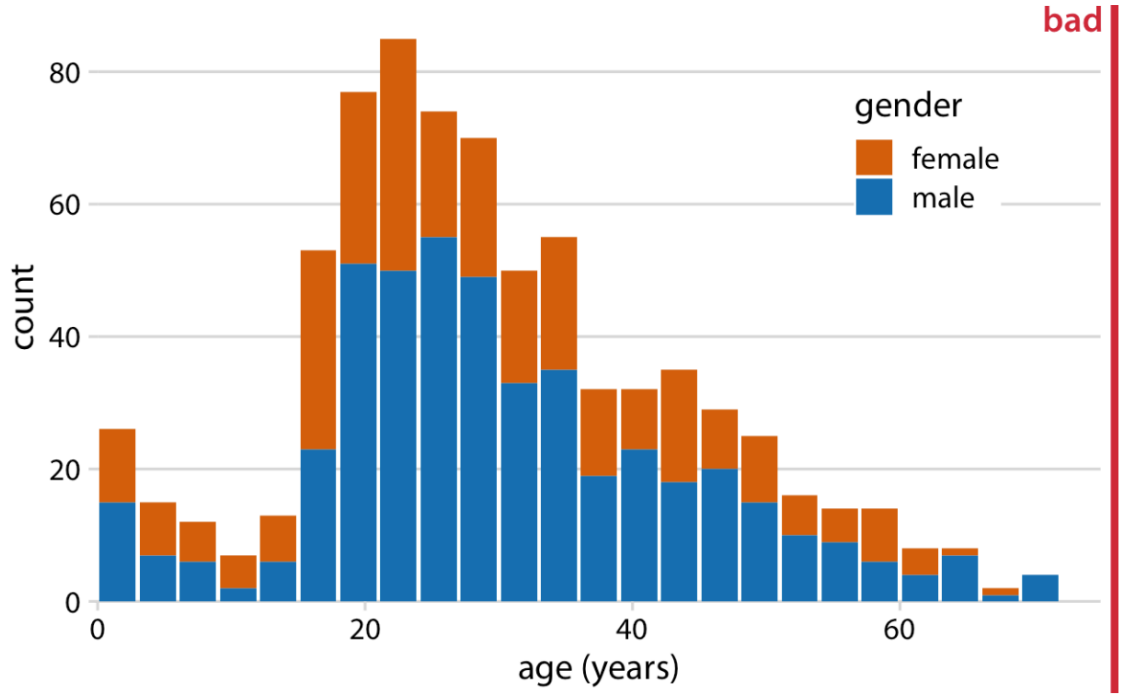
One numeric column, one categorical column. The box shows every car; the bar shows one number per category. Price axis is log.



TODAY'S LECTURE

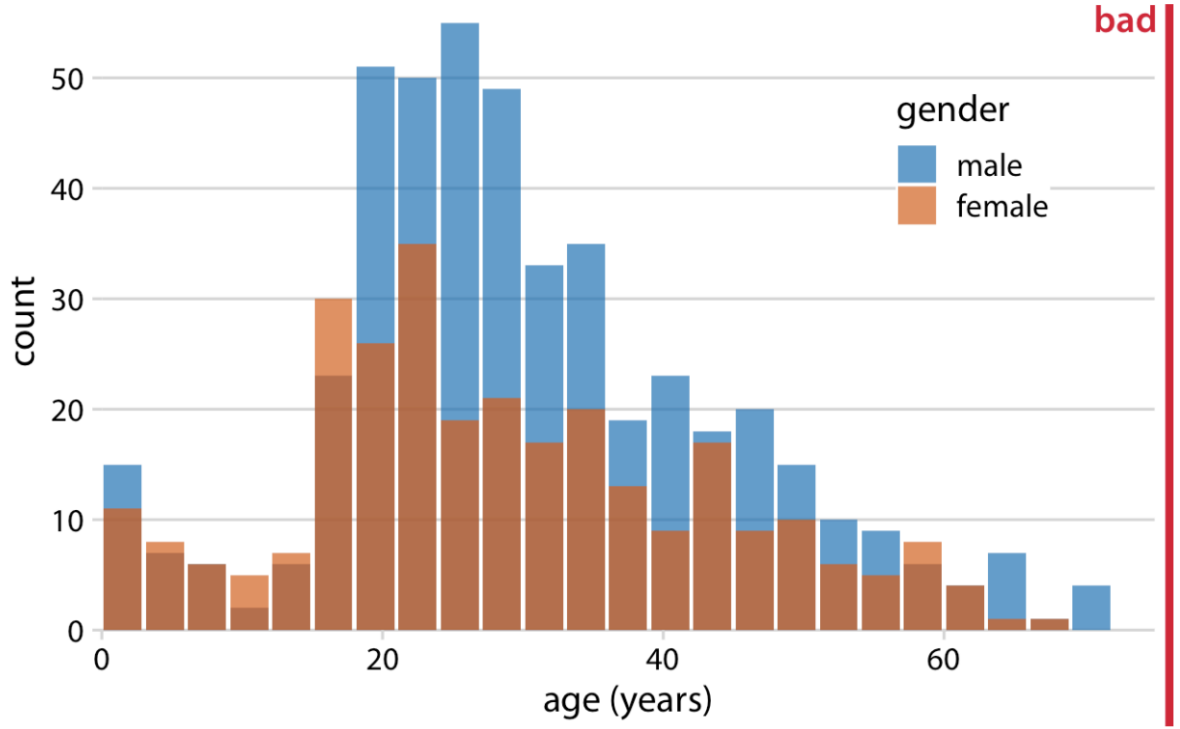
- Review Last Session
- EDA Task
- Uncertainty Plot
- **Visualizing Multiple Distributions**

Visualize Multiple Distribution



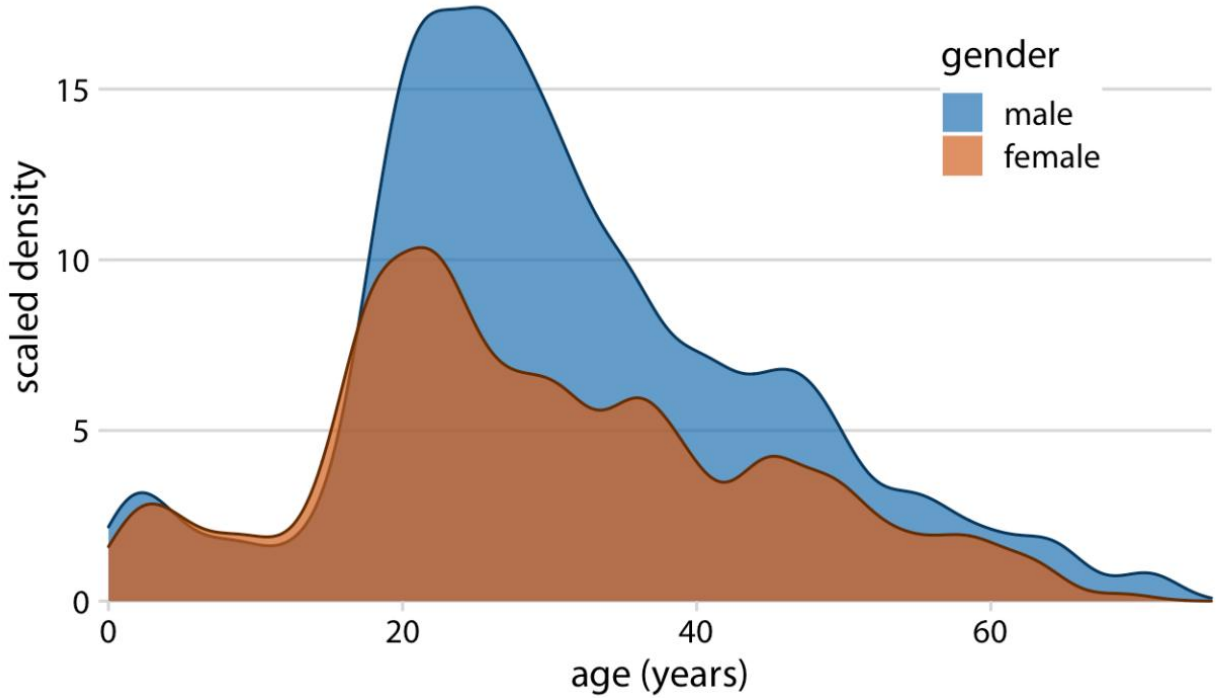
- Figure 7-6. Histogram of the ages of Titanic passengers separated by gender. This figure has been labeled as “bad” because stacked histograms are easily confused with overlapping histograms (see Figure 7-7). In addition, the heights of the bars representing female passengers cannot easily be compared to each other.

Visualize Multiple Distribution



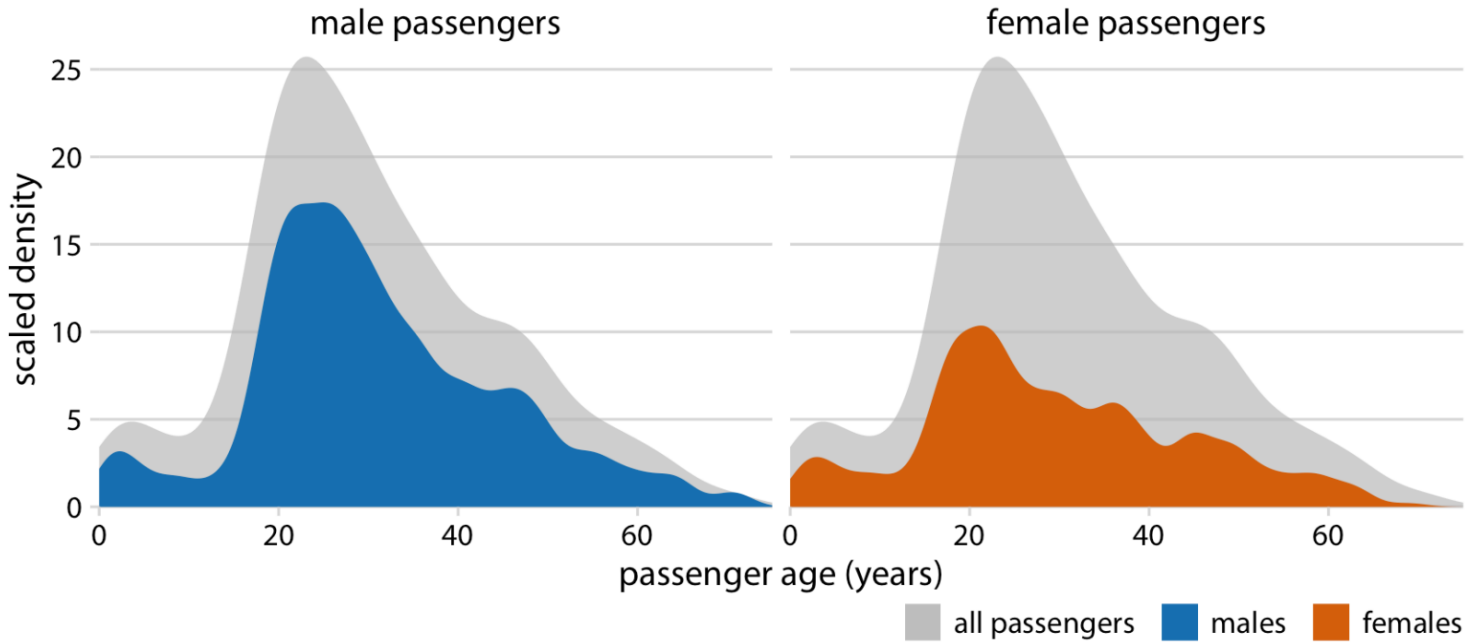
- Figure 7-7. Age distributions of male and female Titanic passengers, shown as two overlapping histograms. This figure has been labeled as “bad” because there is no clear visual indication that all blue bars start at a count of 0.

Visualize Multiple Distribution



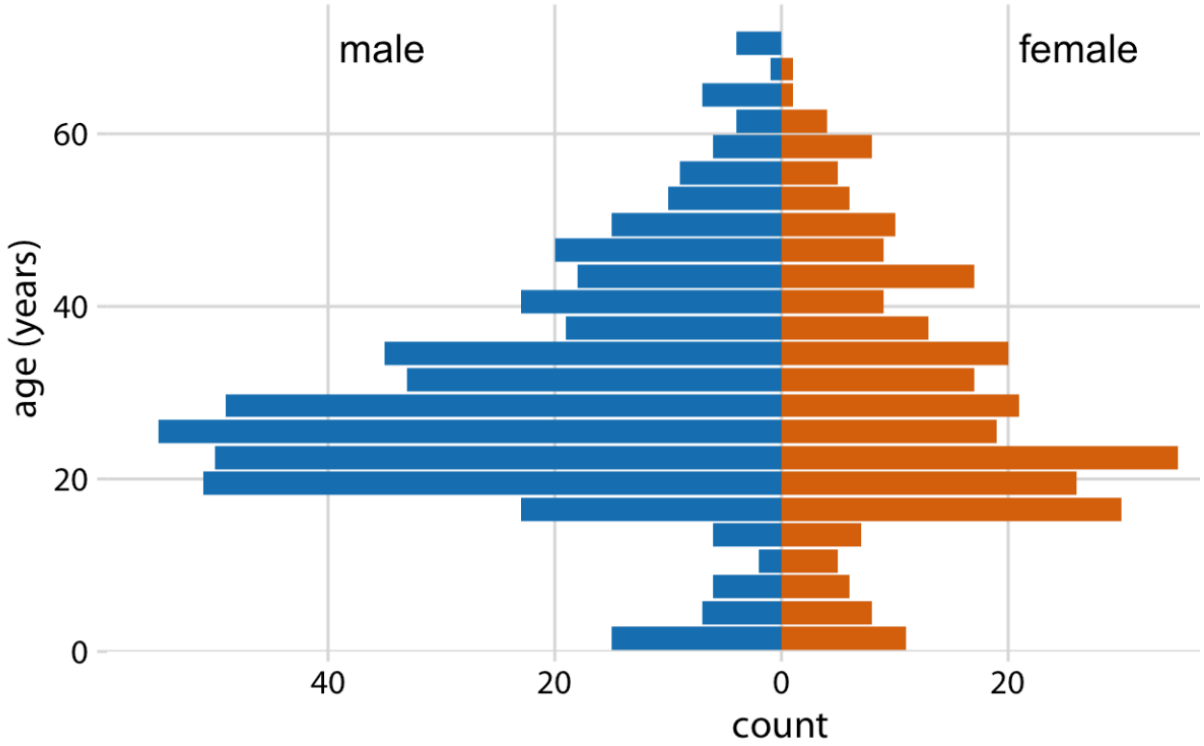
- Figure 7-8. Density estimates of the ages of male and female Titanic passengers. To highlight that there were more male than female passengers, the density curves were scaled such that the area under each curve corresponds to the total number of male and female passengers with known age (468 and 288 respectively).

Visualize Multiple Distribution



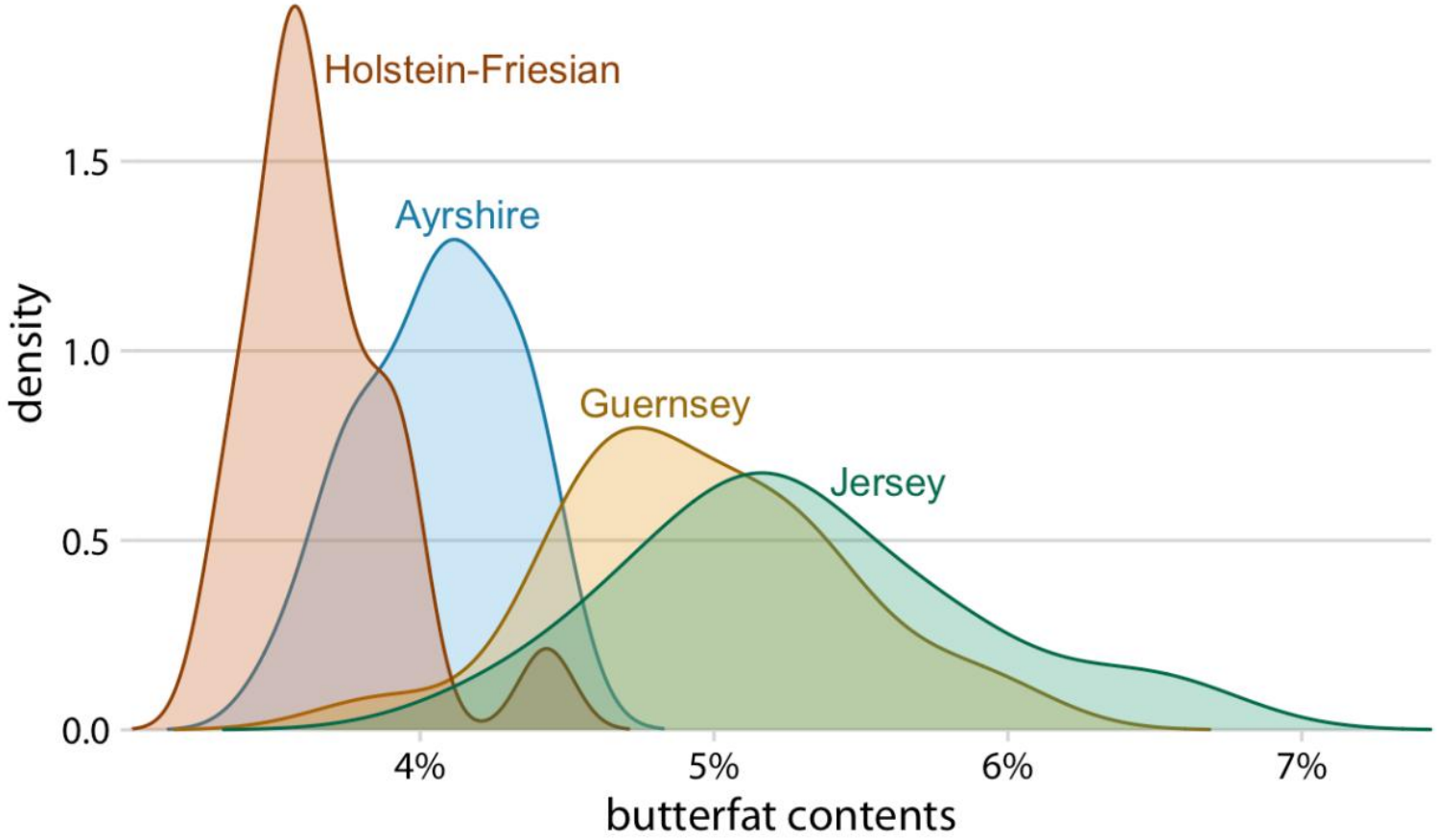
- Figure 7-9. Age distributions of male and female Titanic passengers, shown as proportions of the total number of passengers. The colored areas show the density estimates of the ages of male and female passengers, respectively, and the gray areas show the overall passenger age distribution.

Visualize Multiple Distribution



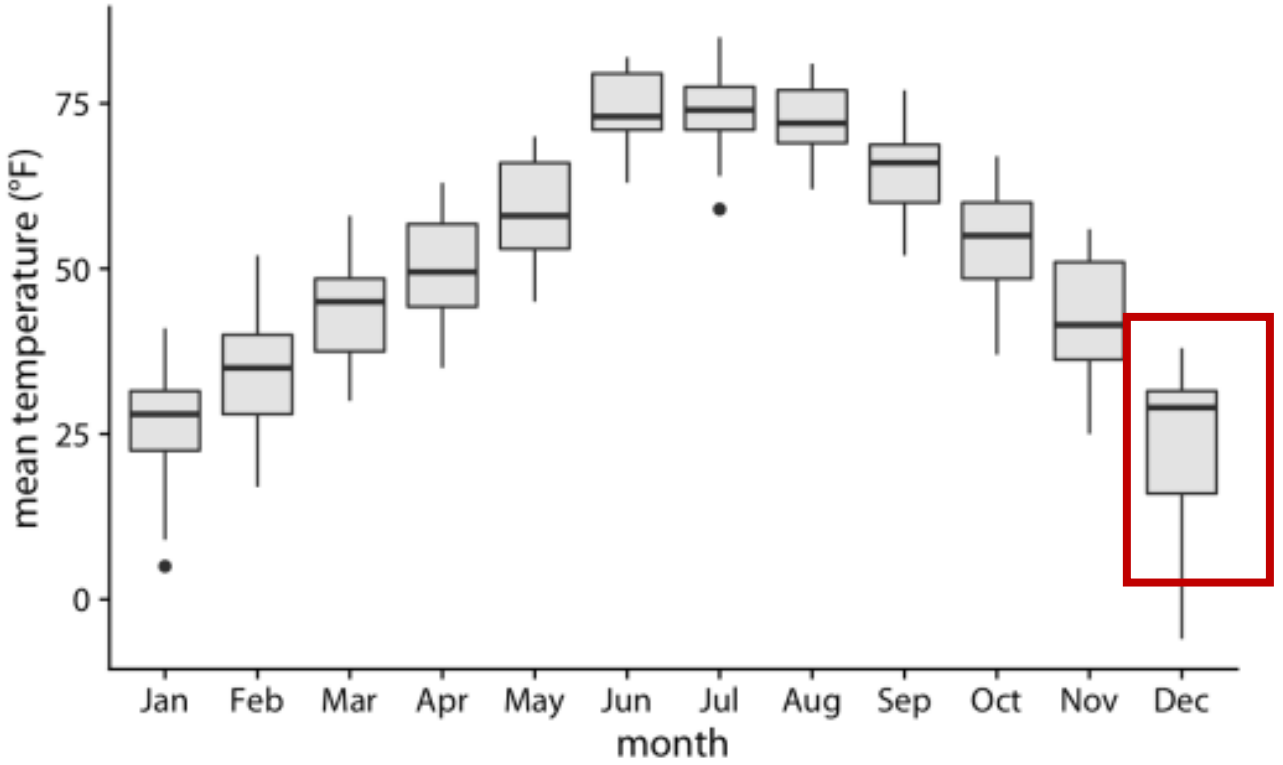
- Figure 7-10. The age distributions of male and female Titanic passengers visualized as an age pyramid.

Visualize Multiple Distribution



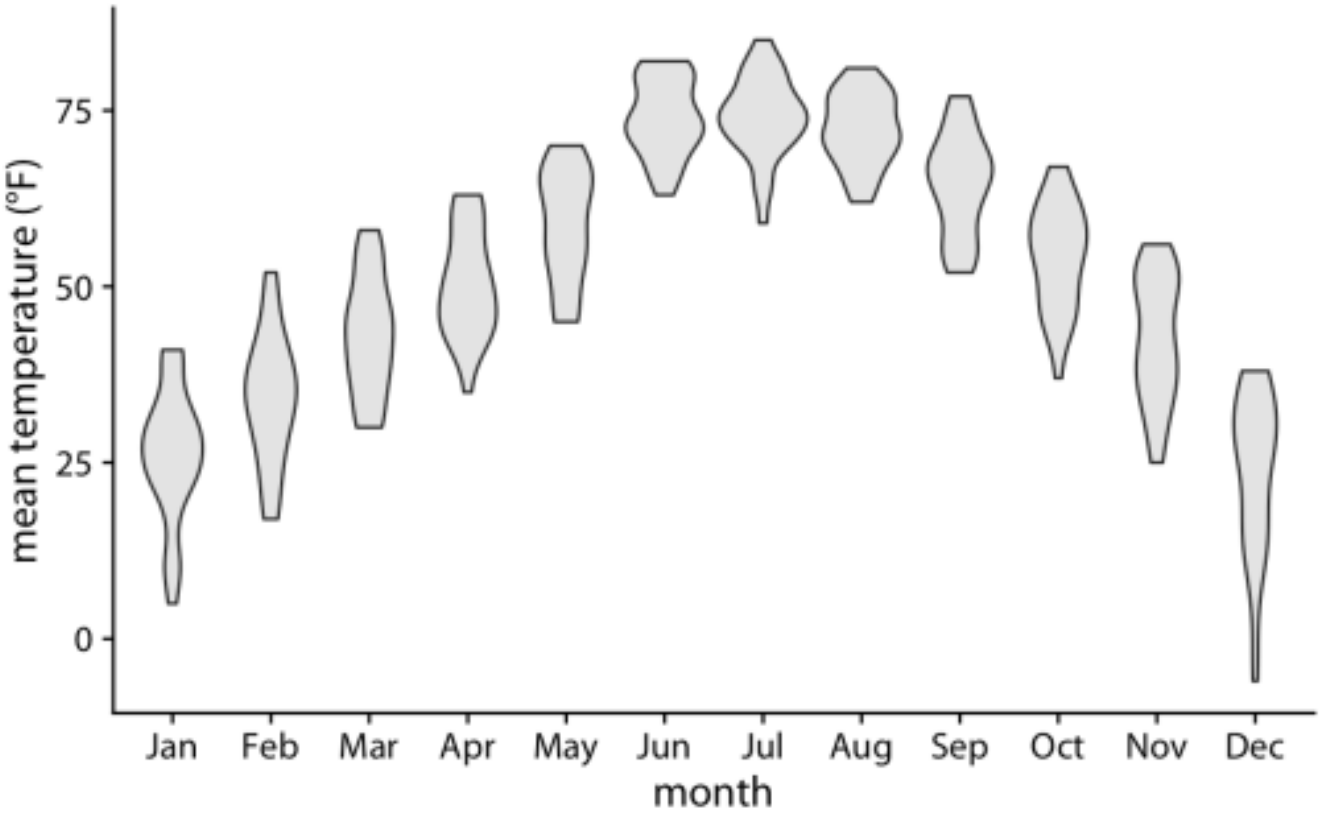
- Figure 7-11. Density estimates of the butterfat percentage in the milk of four cattle breeds. Data source: Canadian Record of Performance for Purebred Dairy Cattle

Visualizing Many Distributions At Once



- Figure 9-3. Mean daily temperatures in Lincoln, NE, visualized as boxplots. Data source: weather underground

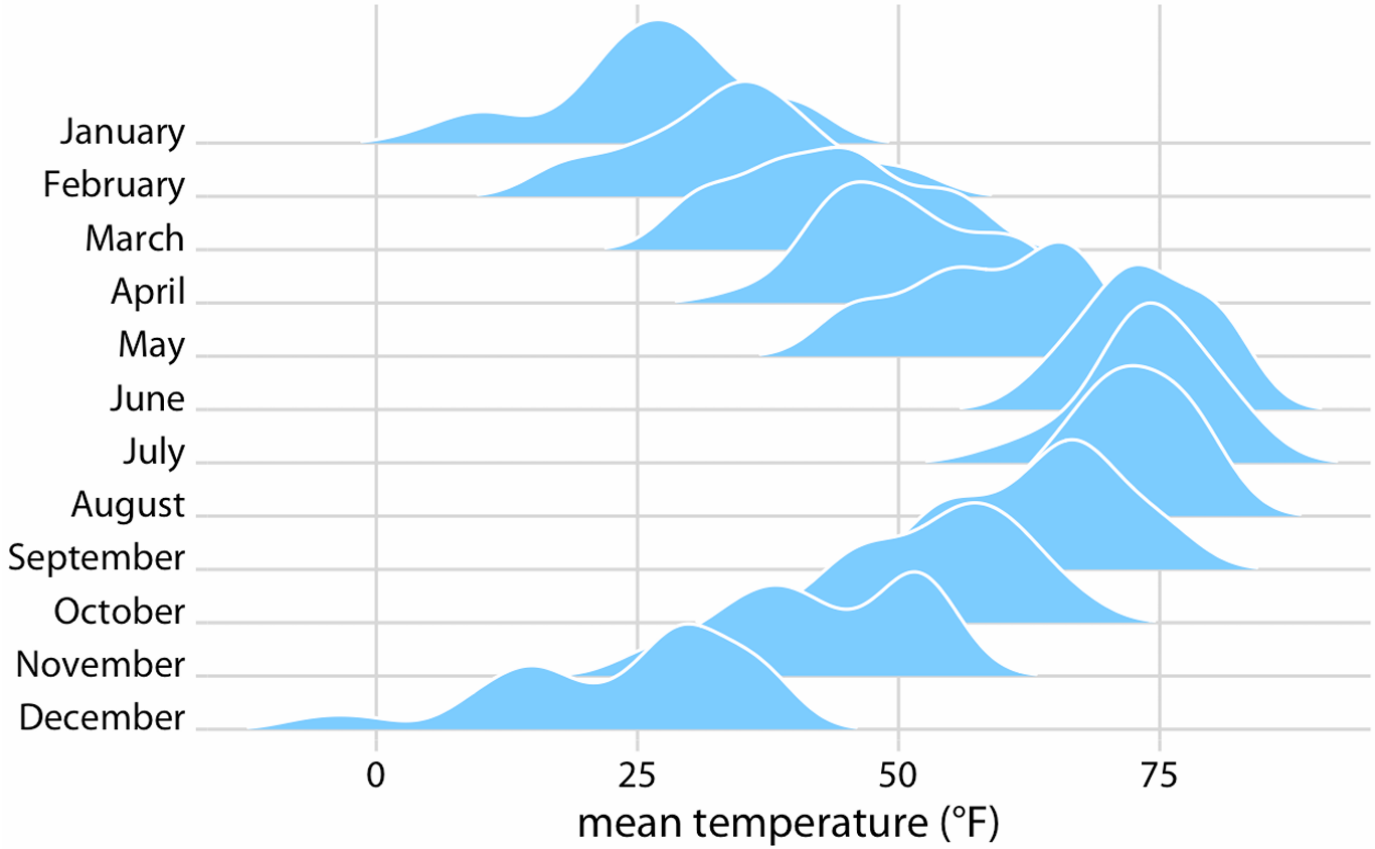
Visualizing Many Distributions At Once



- Figure 9-5. Mean daily temperatures in Lincoln, NE, visualized as violin plots.

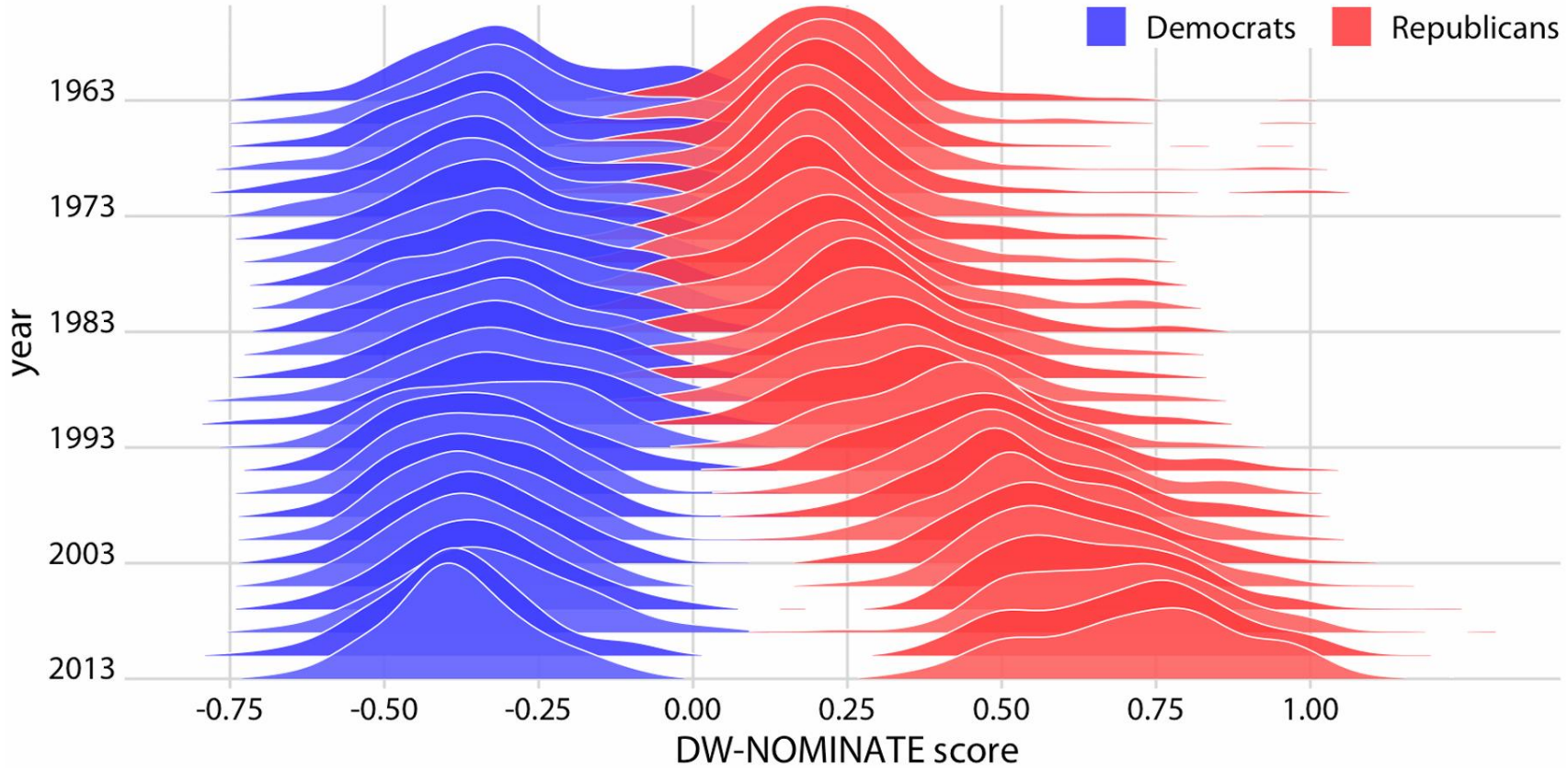
Data source: Weather Underground.

Visualizing Many Distributions At Once



- Figure 9-9. Temperatures in Lincoln, NE, in 2016, visualized as a ridgeline plot. For each month, we show the distribution of daily mean temperatures measured in Fahrenheit. Data source: Weather Underground

Visualizing Many Distributions At Once

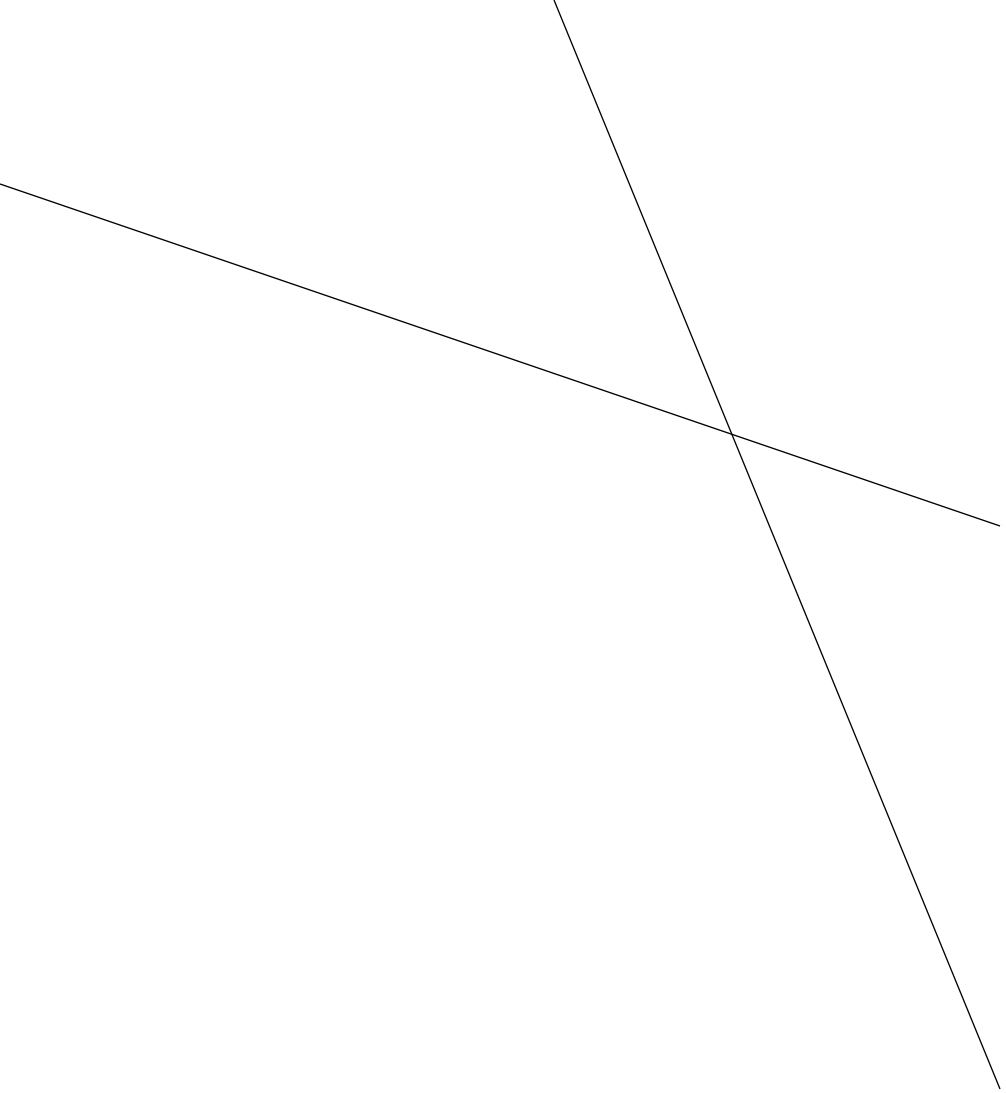


- Figure 9-12. Voting patterns in the US House of Representatives have become increasingly polarized. DW-NOMINATE scores are frequently used to compare voting patterns of representatives between parties and over time.

Before Next Week: Install Tableau Desktop

- **Next class (Sept 23): Tableau I**
  - Data connection, calculated fields, core chart types. Please come with Tableau installed on your laptop.
- **Direct download: Tableau Desktop Free Edition (Windows / macOS):**

  https://www.tableau.com/products/desktop



THANK YOU